MATH 310: Abstract Linear Algebra

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Contents

Introduction **iii**

1 Vector Spaces **1**
1.1 Vector Spaces 1
1.2 Subspaces 4
1.3 Linear Combinations 5
1.4 Linear Dependence and Linear Independence 7
1.5 Bases and Dimension 8

2 Linear Transformations and Matrices **13**
2.1 Linear Transformations, Null spaces and Ranges 13
2.2 The Matrix Representation of a Linear Transformation 15
2.3 Composition of Linear Transformations and Matrix Multiplication 17
2.4 Invertability and Isomorphisms 18
2.5 The Change of Coordinate Matrix 21
2.6 Dual Spaces 22

4 Determinants **26**
4.1 Some Reminders about Determinants 26

5 Diagonalization **27**
5.1 Eigenvalues and Eigenvectors 27
5.2 Diagonalizability 30
5.3 Direct Sums 35
5.4 Invariant subspaces and Cayley-Hamilton Theorem 36

7 Canonical Forms **40**
7.1 The Jordan Canonical Form 40

D Complex Numbers **45**

G Topics Covered in Final Exam **48**

Introduction

Previous classes in Linear Algebra, namely MATH 221 and MATH 307 dealt with $\mathbb{R}^n$ or $\mathbb{C}^n$ but this course will be more abstract. It is based off of *Linear Algebra*, Fifth edition, by Stephen H. Friedberg, Arnold J. Insel and Lawrence E. Spence.
Even though we will be working in more abstract vector spaces than just $\mathbb{R}^n$, they are usually isomorphic. So the names will be different but the structure will be the same.

Chapter 1

Vector Spaces

1.1 Vector Spaces

Definition. Denote the field $\mathbb{F}$ to be one of $\mathbb{Q}, \mathbb{R}, \mathbb{C}$.

A **vector space** V over $\mathbb{F}$ is a non-empty set on which two operations (**addition** and **scalar multiplication**) are defined so that for each pair of elements x, y in V there is a unique element $x + y$ in V and for each element a in $\mathbb{F}$ and each element x in V there is a unique element ax in V such that the following conditions hold.

(VS 1) For all x, y in V, $x + y = y + x$ (commutativity of addition).

(VS 2) For all x, y, z in V, $(x + y) + z = x + (y + z)$ (associativity of addition).

(VS 3) There exists an element in V denoted $\mathbb{O}$ such that $\mathbb{O} + x = x$ for each x in V.

(VS 4) For each element x in V there is an element y in V such that $x + y = 0$.

(VS 5) For each element x in V, $1x = x$.

(VS 6) For each pair of element a, b in $\mathbb{F}$ and each element x in V, $(ab)x = a(bx)$.

(VS 7) For each element a in $\mathbb{F}$ and each pair of elements x, y in V, $a(x + y) = ax + ay$.

(VS 8) For each pair of elements a, b in $\mathbb{F}$ and each element x in V, $(a + b)x = ax + bx$.

The elements $x+y$ and ax are called the **sum** of x and y and **product** of a and x, respectively.

Example. Let $\mathbb{F}^n = \{(a_1, a_2, ..., a_n) | a_1, a_2, ..., a_n \in \mathbb{F}\}$. For $x = (x_1, x_2, ..., x_n)$ and $y = (y_1, y_2, ..., y_n)$, we define

$$x + y \coloneqq (x_1 + y_1, x_2 + y_2, ..., x_n + y_n) \text{ and } cx \coloneqq (cx_1, cx_2, ..., cx_n)$$

Then $(\mathbb{F}^n, +, \cdot)$ is a vector space over $\mathbb{F}$.

Example. If $A = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix}$, then $A_{i,j} \coloneqq a_{i,j}$ for $1 \leq i \leq n$ and $1 \leq j \leq m$.
Let $M_{m\times n}(\mathbb{F})$ be the set of all $m \times n$ matrices with entries from $\mathbb{F}$ with addition and scalar multiplication as follows:
For A, B in $M_{m\times n}(\mathbb{F})$ and c in $\mathbb{F}$

$$(A + B)_{i,j} = A_{i,j} + B_{i,j} \text{ and } (cA)_{i,j} = cA_{i,j}.$$

Then $M_{m\times n}(\mathbb{F})$ is a vector space over $\mathbb{F}$ and $\mathbb{O}$ is the matrix with all 0 entries.

Example. Let S be a non-empty set and let $\mathcal{F}(S, \mathbb{F})$ denote the set of all functions from S to $\mathbb{F}$.
For $f, g \in \mathcal{F}(S, \mathbb{F})$ and $c \in \mathbb{F}$ we define:

$$(f + g)(x) \coloneqq f(x) + g(x) \text{ and } (cf)(x) \coloneqq c[f(x)]$$

for each $x \in S$. Then $\mathcal{F}(S, \mathbb{F})$ is also a vector space over $\mathbb{F}$ where $\mathbb{O}(x) \coloneqq 0$ for all $x \in S$, with the usual 0 of $\mathbb{F}$.

Example. Consider

$$P(\mathbb{F}) = \{a_n x^n + \cdots + a_1 x + a_0 | a_0, \dots, a_n \in \mathbb{F}, n \text{ is a non-negative integer}\}.$$

Let

$$f = a_n x^n + \cdots + a_1 x + a_0 \in P(\mathbb{F}), a_n \neq 0,$$

and

$$g = b_m x^m + \cdots + b_1 x + b_0 \in P(\mathbb{F}), b_m \neq 0,$$

where we assume $m \leq n$, in which case we can write g as follows:

$$g = b_n x^n + \cdots + b_m x^m + \dots b_1 x + b_0$$

with $b_n = \cdots = b_{m+1} = 0$.
Then we can define $f + g$ as follows:

$$f + g = (a_n + b_n)x^n + \dots (a_1 + b_1)x + (a_0 + b_0)$$

and

$$cf = (ca_n)x^n + \cdots + (ca_1)x + ca_0.$$

Then $P(\mathbb{F})$ is a vector space over $\mathbb{F}$.
Remark. $P_n(\mathbb{F}) = \{f \in P(\mathbb{F}) | \deg(f) \leq n\}$ is also a vector space over $\mathbb{F}$.

Example (Not a vector field). Let $\mathbb{R}^+ = \{x | x \in \mathbb{R} \text{ and } x > 0\}$.
We define addition $\oplus$ and scalar multiplication $\odot$ by the following:

$$x \oplus y \coloneqq xy$$
$$a \odot x \coloneqq x^{2a}$$

where $x, y \in \mathbb{R}^+, a \in \mathbb{R}$.
However $1 \odot x = x^2 \neq x$ for any $x \neq 1, x \in \mathbb{R}^+$, so (VS 5) is violated.

Example (Not a vector field). Let $\mathbb{R}^+ = \{x | x \in \mathbb{R} \text{ and } x > 0\}$.
We define addition $\oplus$ and scalar multiplication $\odot$ by the following:

$$x \oplus y \coloneqq xy$$
$$a \odot x \coloneqq x^a$$

where $x, y \in \mathbb{R}^+, a \in \mathbb{R}$. Then $\mathbb{R}^+$ is a vector space over $\mathbb{R}$ with respect to the operations defined above.

Let us prove some facts about vector addition.

Theorem 1.1 (Cancellation Law for Vector Addition). If x, y and z are vectors in a vector space V such that $x + z = y + z$ then $x = y$.

Proof. By (VS 4) there exists a vector v in V such that $z + v = \mathbb{O}$.
Thus by (VS 2) and (VS 3),

$$\begin{aligned} x &= x + \mathbb{O} = x + (z + v) = (x + z) + v \\ &= (y + z) + v = y + (z + v) = y + \mathbb{O} = y. \end{aligned}$$

□

Corollary 1.1.1. The vector $\mathbb{O}$ in (VS 3) is unique and is called the **zero vector**.

Proof. Suppose the vector $\mathbb{O}'$ has property (VS 3) then $\mathbb{O}' = \mathbb{O}' + \mathbb{O} = \mathbb{O}$. □

Corollary 1.1.2. The vector y in (VS 4) is unique and is called the **additive inverse** denoted $-x$.

Proof. Suppose y' has property (VS 4) then $x + y' = x + y = 0$, hence by the cancellation law of vector addition $y' = y$. □

The following theorem contains elementary properties of scalar multiplication.

Theorem 1.2. In any vector space V, the following statements are true:

(a) $0x = \mathbb{O}$ for each $x \in V$.

(b) $(-a)x = -(ax) = a(-x)$ for each $a \in \mathbb{F}$ and each $x \in V$.

(c) $a\mathbb{O} = \mathbb{O}$ for each $a \in \mathbb{F}$.

Proof. Let $a \in \mathbb{F}$ and $x \in V$.

(a) By (VS 8), (VS 3) and (VS 1), $0x + 0x = (0 + 0)x = 0x = 0x + \mathbb{O} = \mathbb{O} + 0x$. Hence by Theorem 1.9, $0x = \mathbb{O}$.

(b) We know $ax + (-(ax)) = \mathbb{O}$, then by (VS 8), $ax + (-a)x = (a + (-a))x = 0x = \mathbb{O}$ so by Corollary 1.9.2, $-(ax) = (-a)x$.

    Then we also have $(-a)x = (a(-1)x) = a(-x)$.

(c) By (VS 6) and (a), $a\mathbb{O} = a(0x) = (a0)x = 0x = \mathbb{O}$ for each $x \in V$.

□

Example. Let $S = \{(a_1, a_2) | a_1, a_2 \in \mathbb{R}\}$
For $(a_1, a_2), (b_1, b_2) \in S, c \in \mathbb{R}$ we define

$$(a_1 + a_2) + (b_1 + b_2) = (a_1 + b_1, a_2 - b_2)$$
$$c(a_1, a_2) = (ca_1, ca_2)$$

Then S is not a vector space because $(0, 1) + (0, 2) = (0, -1)$ and $(0, 2) + (0, 1) = (0, 1)$ which implies $(0, 1) + (0, 2) \neq (0, 2) + (0, 1)$ i.e. (VS 1) fails.

Example. Let $S = \{(a_1, a_2) | a_1, a_2 \in \mathbb{R}\}$
For $(a_1, a_2), (b_1, b_2) \in S, c \in \mathbb{R}$ we define

$$(a_1 + a_2) + (b_1 + b_2) = (a_1 + b_1, 0)$$
$$c(a_1, a_2) = (ca_1, 0)$$

Then S is not a vector space because $1(a_1, a_2) = (a_1, 0) \neq (a_1, a_2)$, i.e. (VS 5) fails.

Lemma. If V is a vector space over $\mathbb{F}$, $v \in V$ and $c \in \mathbb{F}$ then $cv = \mathbb{O}$ implies either $c = 0$ or $v = \mathbb{O}$.

Proof. Let $c \in \mathbb{F}$ and $v \in V$.
If $c = 0$ then we are done.
If $c \neq 0$ then we have $v = 1v = \left(\frac{1}{c}c\right) v = \frac{1}{c}\mathbb{O} = \mathbb{O}$. □

Lemma. If V is a vector space over $\mathbb{F}$ and $V \neq \{\mathbb{O}\}$, then V is an infinite set.

Proof. Let $v \in V$. If $c, d \in \mathbb{F}$ and $c \neq d$ then $cv \neq dv$ for $v \neq \mathbb{O}$. In fact, if $cv = dv$ then $cv - dv = \mathbb{O}$ implies $(c - d)v = 0v = \mathbb{O}$. □

1.2 Subspaces

Definition. A subset W of a vector space V over $\mathbb{F}$ is called a subspace of V if W is a vector space over $\mathbb{F}$ with the operations of addition and scalar multiplication defined on V.

Example. If V is a vector space over $\mathbb{F}$ then $\{\mathbb{O}\}$ and V are subspaces of V.

Note. Any subspace not equal to one of the above is called **non-trivial**.

Theorem 1.3. Let V be a vector space and W a subset of V. Then W is a vector space if and only if the following three conditions hold for the operations defined in V.

(a) $0 \in W$

(b) $x + y \in W$ wherever $x \in W$ and $y \in W$

(c) $cx \in W$ whenever $c \in \mathbb{F}$ and $x \in W$

Proof. Let V be a vector space and $W \subset V$.
($\Rightarrow$) If W is a subspace of V, then W is a vector space with the operations defined on V. Hence (b) and (c) will hold and there exists a vector $\mathbb{O}'$ such that $x + \mathbb{O}' = x$ for each $x \in W$. But also $x + \mathbb{O} = x$, and thus $\mathbb{O}' = \mathbb{O}$ by Theorem 1.1. So (a) holds.
($\Leftarrow$) Now if (a), (b) and (c) hold, we know Properties (VS 1), (VS 2), (VS 5), (VS 6), (VS 7) and (VS 8) hold for all vectors in V, so they hold for all vectors in W. Then by property (c), $(-1)x = -x \in W$, so (VS 4) holds. Hence W is a vector space. □

Remark. The following statement is equivalent to Theorem (1.3).
Let W be a non-empty subset of a vector space V. Then W is a vector subspace of V if and only if the following two conditions hold for the operations defined in V.

(a) $x - y \in W$ whenever $x \in W$ and $y \in W$.

(b) $cx \in W$ whenever $c \in \mathbb{F}$ and $x \in W$

Example. Consider $V = M_{n\times n}(\mathbb{F}) = \{A | A \text{ is a } n \times n \text{ matrix with entries in } \mathbb{F}\}$.
Let $W = \{A | A \in M_{n\times n}, \operatorname{tr}(A) = 0\}$. Then W is a subspace of V.

Proof. Clearly $W \neq \emptyset$ because $\mathbb{O} \in W$.
If $x \in W$ and $y \in W$ then $\operatorname{tr}(x + y) = \operatorname{tr}(x) + \operatorname{tr}(y) = 0$.
If $x \in W$ and $c \in \mathbb{F}$ then $\operatorname{tr}(cx) = c\operatorname{tr}(x) = 0$.
So $\mathbb{O} \in W$, $x + y \in W$ and $cx \in W$. By theorem (1.3), W is a vector subspace. □

Theorem 1.4. Any intersection of subspaces of a vector space V is a subspace of V.

Proof. Let $\mathcal{C}$ be a collection of subspaces of V, and let W denote the intersection of the subspaces in $\mathcal{C}$. Since every subspace contains the zero vector, $\mathbb{O} \in W$. Let $a \in \mathbb{F}$ and $x, y \in W$. Then x and y are contained in each subspace in $\mathcal{C}$. Because each subspace in $\mathcal{C}$ is closed under addition and scalar multiplication, it follows that $x + y$ and ax are contained in each subspace in $\mathcal{C}$. Hence $x + y$ and ax are also contained in W, so that W is a subspace of V by Theorem 1.3. □

1.3 Linear Combinations

Definition. Let V be a vector space over $\mathbb{F}$ and S a non empty subset of V. A vector $v \in V$ is called a **linear combination of vectors** in S if there exists a finite number of vectors $u_1, \ldots, u_n \in S$ and scalars $a_1, \ldots, a_n \in \mathbb{F}$ such that

$$v = a_1u_1 + \cdots + a_n + u_n.$$

In this case we also say v is a linear combination of $u_1, \ldots, u_n$ and $a_1, \ldots, a_n$ are called the **coefficients** of the linear combination.

Definition. Let S be a non-empty subset of a vector space. The **span** of S, denoted $\text{span}(S)$, is the set consisting of all linear combinations of vectors in S. For convenience, we define $\text{span}(\emptyset) = \{\mathbb{O}\}$.

Theorem 1.5. The span of any subset S of a vector space is subspace of V. Moreover, any subspace of V that contains S must contain the span of S.

Proof. The result is immediate if $S = \emptyset$ because $\text{span}(\emptyset) = \{\mathbb{O}\}$, which is a subspace that contains S and is contained in any vector space V.
If $S \neq \emptyset$, then S contains a vector z. So $0z = \mathbb{O}$ is in $\text{span}(S)$. Let $x, y \in \text{span}(S)$. Then there exist vectors $u_1, u_2, \ldots, u_m,\ v_1, v_2, \ldots, v_n$ in S and scalars $a_1, a_2, \ldots, a_m,\ b_1, b_2, \ldots, b_n$ be such that

$$x = a_1u_1 + a_2u_2 + \cdots + a_mu_m \text{ and } y = b_1v_1 + b_2v_2 + \ldots b_nv_m.$$

Then

$$x + y = a_1u_1 + a_2u_2 + \cdots + a_mu_m + b_1v_1 + b_2v_2 + \ldots b_nv_m$$

and, for any scalar c,

$$cx = (ca_1)u_1 + (ca_2)u_2 + \cdots + (ca_m)u_m$$

are linear combinations of the vectors in S; so $x + y$ and cx are in $\text{span}(S)$. Thus $\text{span}(S)$ is a subspace of V. Furthermore, if $v \in S$, then $v = 1 \cdot v \in \text{span}(S)$; so $\text{span}(S)$ contains S.
Now let W denote any subspace of V that contains S. If $w \in \text{span}(S)$, then w has the form $w = c_1w_1 + c_2w_2 + \ldots c_kw_k$ for some vectors $w_1, w_2, \ldots, w_k$ in S and some scalars $c_1, c_2, \ldots c_k$. Since $S \subseteq W$, we have $w_1, w_2, \ldots, w_k \in W$. Therefore $w = c_1w_1 + \ldots c_kw_k$ is in W. Because w was an arbitrary vector in $\text{span}(S)$ and belongs to W, it follows that $\text{span}(S) \subseteq W$. $\square$

Definition. A subset S of a vector space S **spans** V if $\text{span}(S) = V$. In this case we also say that the vectors of S span V.

Example. The vectors $(1, 1, 0), (1, 0, 1)$ and $(0, 1, 1)$ generate $\mathbb{R}^3$ since any arbitrary vector (a_1, a_2, a_3) in $\mathbb{R}^3$ is a linear combination of the three given vectors. In factor the scalars r, s and t for which

$$r(1, 1, 0) + s(1, 0, 1) + t(0, 1, 1) = (a_1, a_2, a_3)$$

are

$$r = \frac{1}{2}(a_1 + a_2 + a_3), s = \frac{1}{2}(a_1 - a_2 + a_3), \text{ and } t = \frac{1}{2}(-a_1 + a_2 + a_3).$$

1.4 Linear Dependence and Linear Independence

Definition. A subset S of a vector space V is called **linearly dependent** if there exist a finite number of distinct vectors $u_1, \ldots, u_n$ in S and scalars $a_1, \ldots, a_n$ in $\mathbb{F}$ such that $a_1, \ldots, a_n$ are not all 0 and

$$a_1u_1 + \cdots + a_nu_n = \mathbb{O}.$$

Definition. A subset S of a vector space V is called **linearly independent** if it is not linearly dependent.

Remark. The empty set is linearly independent, for linearly independent sets must be non-empty.
A set consisting of a single vector is linearly independent.
A set is linearly independent iff the only representations of $\mathbb{O}$ as linear combinations of its vectors are the trivial representations.

Theorem 1.6. Let V be a vector space and $S_1 \subseteq S_2 \subseteq V$. If S_1 is linearly dependent then S_2 is linearly dependent.

Proof. Since S_1 is linearly dependent, there exist vectors $u_1, \ldots, u_n$ in S_1 and scalars $a_1, \ldots, a_n$ in $\mathbb{F}$ which are not all 0, such that

$$a_1u_1 + \cdots + a_nu_n = \mathbb{O}. \tag{*}$$

Then since $S_1 \subseteq S_2, u_1, \ldots, u_n$ are in S_2. Hence S_2 is linearly dependent by (*). □

Corollary 1.6.1. Let V be a vector space and let $S_1 \subseteq S_2 \subseteq V$. If S_2 is linearly independent, then S_1 is linearly independent.

Proof. Assume for a contradiction that S_1 is linearly dependent. Then from Theorem (1.6), it follows that S_2 is linearly dependent which is a contradiction. Hence S_1 must be linearly independent. □

Theorem 1.7. Let S be a linearly independent subset of a vector space V. Let v be a vector in V that is not in S.
Then $S \cup \{v\}$ is linearly dependent iff $v \in \text{span}(S)$.

Proof. ($\Rightarrow$) Assume that $S \cup \{v\}$ is linearly dependent. Then there are vectors $u_1, \ldots, u_n$ in S and scalars $a_1, \ldots, a_n, b$ in $\mathbb{F}$ that are not all 0, such that

$$a_1u_1 + \ldots a_nu_n + bv = \mathbb{O}. \tag{*}$$

Then since S is linearly independent, we know b must not be 0. Hence by (*) we have

$$v = \frac{1}{b}((-a_1)u_1 + \cdots + (-a_n)u_n)$$

Since v can be written as a linear combination of vectors in S, $v \in \text{span}(S)$.

($\Leftarrow$) Now assume that $v \in \text{span}(S)$. Then there are vectors $u_1, \ldots, u_n$ in S and scalars $a_1, \ldots, a_n$ in $\mathbb{F}$ such that

$$v = a_1 u_1 + \ldots a_n u_n$$

Then

$$v + (-a_1)u_1 + \cdots + (-a_n)u_n = \mathbb{O}$$

Hence the subset $u_1, \ldots, u_n, v$ of $S \cup \{v\}$ is linearly dependent. And by Theorem 1.6, $S \cup \{v\}$ is linearly dependent. $\square$

1.5 Bases and Dimension

Definition. A **basis** β of a vector space V is a linearly independent subset of V that spans V. If β is a basis for V, we also say the vector of β form a basis of V.

Example. Recall that $\text{span}(\emptyset) = \mathbb{O}$ so $\emptyset$ forms a basis for the zero vector space.

Example. In $\mathbb{F}^n$, let $e_1 = (1, 0, \ldots, 0), e_2 = (0, 1, \ldots, 0), \ldots, e_n = (0, 0, \ldots, 1)$. Then $\bigcup_{i=1}^n \{e_i\}$ forms a basis of $\mathbb{F}^n$ and is called the **standard basis** for $\mathbb{F}^n$.

Example. Consider $P(\mathbb{F})$. Then the set $\{1, x, x^2, \ldots, x^n, \ldots\}$ forms a basis of $P(\mathbb{F})$.

Note. This tells us a basis does not have to be a finite set.

Theorem 1.8. Let V be a vector space and $\beta = \{u_1, \ldots, u_n\}$, then β is a basis for V iff every vector of V can be uniquely expressed as a linear combination of vectors of β, that is, $v \in V$ can be written in the form

$$v = a_1 u_1 + \cdots + a_n u_n$$

for unique scalars $a_1, \ldots, a_n$.

Proof. ($\Rightarrow$) Suppose β is a basis for V. Then $v \in V = \text{span}(\beta)$ and

$$v = a_1 u_1 + \cdots + a_n u_n \tag{1}$$

where $u_1, \ldots, u_n$ are vectors in β and $a_1, \ldots, a_n$ are scalars in $\mathbb{F}$.
Now assume for a contradiction that there are scalars $b_1, \ldots, b_n$ in $\mathbb{F}$ such that not all $a_1 = b_1$ and

$$v = b_1 u_1 + \cdots + b_n u_n. \tag{2}$$

Then by combining (1) and (2), we have

$$(a_1 - b_1)u_1 + \cdots + (a_n - b_n)u_n = (v - v) = w\mathbb{O}$$

where at least one of $a_1 - b_1 \neq 0$. Hence β is linearly dependent which is a contradiction.
($\Leftarrow$) Assume that every scalar $v \in V$ can be uniquely written as a linear combination of vectors in β. Then $v \in \text{span}(\beta)$.
So, $V \subseteq \text{span}(\beta)$ and β is a spanning set of V.

Now suppose there are vectors $u_1, \ldots, u_n$ in β and scalars $a_1, \ldots, a_n$ in $\mathbb{F}$ such that

$$a_1u_1 + \cdots + a_nu_n = \mathbb{O} = 0u_1 + \cdots + 0u_n.$$

Then by the uniqueness assumption $a_1 = \cdots = a_n = 0$. Hence β is linearly independent. So β is a basis for V. □

Theorem 1.9. If a vector space V is spanned by a finite set S, then some subset of S is a basis of V. Hence V has finite basis.

Proof. If $S = \emptyset$, then $V = \text{span}(S) = \text{span}(\emptyset) = \{\mathbb{O}\}$ in which case $\emptyset$ is a basis for V.
If $S \neq \emptyset$, then for all u_1 in S such that $u_1 \neq \mathbb{O}$ we have two cases:

(1) If $\text{span}(u_1) = V$ in which case we are done because $\{u_1\}$ is a basis for V.

(2) If $\text{span}(u_1) \neq V = \text{span}(S)$, then there exists a u_2 in S such that $u_2 \notin \text{span}(u_1)$ which implies that $\{u_1\} \cup \{u_2\}$ is linearly independent.

    Continue, if possible, choosing vectors $u_1, \ldots, u_k$ such that $\beta = \{u_1, \ldots, u_k\}$ is linearly independent but adjoining to β one more vector in S not in β produces a linearly dependent set (Since S is finite, we know such a β is possible)

    Since β is linearly independent, we only need to prove β is a spanning set, i.e. we want to prove $V = \text{span}(\beta)$.

    Recall $V = \text{span}(S)$, hence its enough to show $S \subseteq \text{span}(\beta)$.

    For all v in S, we have two cases:

    - If $v \in \beta$, then $v \in \text{span}(\beta)$.
    - If $v \notin \beta$, then $\{u_1, \ldots, u_n, v\} = \beta \cup \{v\}$ is linearly dependent by construction of β. Hence there exist scalars $a_1, \ldots, a_k$ and b such that

      $$a_1u_1 + \cdots + a_ku_k + bv = \mathbb{O} \tag{*}$$

      and at least one of the scalars is non=zero.
      Since β is linearly independent, we have to have $b \neq 0$ and solving (*) for v gives

      $$v = \frac{1}{b}(-a_1u_1 - \cdots - u_kv_k).$$

      Hence $v \in \text{span}(\beta)$.

    This proves $S \subseteq \text{span}(\beta)$.

□

Theorem 1.10 (Replacement Theorem). Suppose V is a vector space that is spanned by a set G containing exactly n vectors and L be a linearly independent subset of V containing exactly m vectors.
Then $m \leq n$ and there exists a subset H of G containing exactly $n - m$ vectors such hat $L \cup H$ spans V.

Proof. Let us use mathematical induction on m.
If $m = 0$, then $L = \emptyset$ and $H = G$ works.
Now assume that the theorem holds for some integer $m \geq 0$. We will then prove the theorem for $m + 1$. In this case, let $L = \{v_1, \ldots, v_{m+1}\}$. Since L is linearly independent, then $L \setminus \{v_{m+1}\}$ is linearly independent. By the induction assumption $m \leq n$ and we know there exists a set $\{u_1, \ldots, u_{n-m}\} \subset G$ such that $\{v_1, \ldots, v_m\} \cup \{u_1, \ldots, u_{n-m}\}$ spans V.
Hence

$$v_{m+1} = a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_{n-m} u_{n-m} \tag{*}$$

for some scalars $a_1, \ldots, a_m, b_1, \ldots, b_{n-m}$.
If $n = m$,

$$v_{m+1} = a_1 v_1 + \cdots + a_m v_m$$

which is impossible because L is linearly independent. So $m < n$, i.e. $m + 1 \leq n$. Also, we claim one of b_i is non-zero. Solving (*) for u_1, we get

$$\begin{aligned}
u_1 &= \frac{1}{b_1}(v_{m+1} - a_1 v_1 - \ldots a_m v_m - b_1 u_1 - \cdots - b_n u_n) \\
&\Rightarrow u_1 \in \operatorname{span}(v_1, \ldots, v_m, v_{m+1}, u_2, \ldots, u_{n-m}) \\
&\Rightarrow \{v_1, v_2, \ldots, v_{m+1}\} \cup \{u_1, \ldots, u_{n-m}\} \in \operatorname{span}(\{v_1, v_2, \ldots, v_{m+1}\} \cup \{u_2, \ldots, u_{n-m}\})
\end{aligned}$$

Hence

$$\begin{aligned}
V &= \operatorname{span}(\{v_1, v_2, \ldots, v_m\} \cup \{u_1, \ldots, u_{n-m}\}) \\
&\subseteq \operatorname{span}(\{v_1, \ldots, v_{m+1}\} \cup \{u_1, \ldots, u_{n-m}\}) \\
&\subseteq \operatorname{span}(\{v_1, \ldots, v_{m+1}\} \cup \{u_2, \ldots, u_{n-m}\})
\end{aligned}$$

which implies

$$\operatorname{span}(\{v_1, \ldots, v_{m+1}\} \cup \{u_2, \ldots, u_{n-m}\}) = V.$$

Hence the theorem is true for $m + 1$.

□

Corollary 1.10.1. Let V be a vector space with a finite basis, then every basis for V contains the same number of vectors.

Proof. Suppose that β is a finite basis for V that contains exactly n vectors. Let γ be another basis for V. Assume that γ contains exactly m vectors. By the replacement theorem, we get

$$m \leq n \text{ and } n \leq m$$

So $m = n$. □

Definition. A vector space is called finite-dimensional if it has a basis consisting of finitely many vectors. The unique number of vectors in a basis for V is called the dimension of V and is denoted by $\dim(V) = \dim_{\mathbb{F}}(V)$.

Examples.

- $\dim_{\mathbb{F}}(\mathbb{F}^n) = n$
- $\dim_{\mathbb{F}}(M_{m\times n}(\mathbb{F})) = mn$
- $\dim_{\mathbb{C}}(\mathbb{C}) = 1$
- $\dim_{\mathbb{R}}(\mathbb{C}) = 2$
- $\dim_{\mathbb{F}}(P_n(\mathbb{F})) = n + 1$

Corollary 1.10.2. Let V be a vector space with dimension n.

(a) Any finite spanning set for V has at least n vectors, and a spanning set for V that contains exactly n vectors must be a basis.

(b) Any linearly independent subset of V that contains n vectors must be a basis for V.

(c) Every linearly independent subset of V can be extended to a basis for V.

Proof. We will prove each in turn (proofs of (a) and (b) were not included in lecture).

(a) Let G be a finite spanning set for V. By Theorem (1.9) some subset H of G is a basis for V. Hence H must contain exactly n vectors and G must contain $m \geq n$ vectors. Furthermore if G contains exactly n vectors, then $G = H$ and G is a basis for V.

(b) Let L be a linearly independent subset of V containing exactly n vectors. It follows from replacement theorem that there is a subset H of a basis β containing $m - n = 0$ vectors such that $L \cup H$ spans V. Hence $H = \emptyset$ and L spans V. Since L is linearly independent, L is a basis for V.

(c) Let L be a linearly independent subset of a vector space V with $\dim(V) = n$, containing m vectors. Then by the replacement theorem, we can find a subset H of a basis β of V containing exactly $m - n$ vectors such that $L \cup H$ spans V. Since $L \cup H$ contains n vectors, $L \cup H$ is a basis.

□

Example. Consider $P_2(\mathbb{R})$. Then $\{1, x, x^2\}$ is a basis for $P_2(\mathbb{R})$ and $S = \{x^2 + 3x - 2, 2x^2 + 5x - 3, -x^2 - 4 + 4\}$ is also a basis.

Proof. We will show S is linearly independent. We can set the following matrix and reduce it to reduce row echelon form.

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 5 & -4 \\ 2 & -3 & 4 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then S is a linearly independent set, so S is a basis for $P_2(\mathbb{R})$. □

Theorem 1.11. Let W be a subspace of a finite dimension vector space v. Then W is also finite dimensional and $\dim(W) \leq \dim(V)$.
Moreover if $\dim(W) = \dim(V)$ then $W = V$.

Proof. Let G be a basis for V containing exactly n vectors.
Suppose for a contradiction that $m = \dim(W) > \dim(V) = n$. Then there is a set L consisting of exactly m linearly independent vectors that spans W. However by the replacement theorem $m \leq n$ which is a contradiction.
Now if $m = n$, since L is linearly independent, by Corollary (1.10.2 b), L must be a basis for W and for V. So $W = V$. $\square$

Chapter 2

Linear Transformations and Matrices

2.1 Linear Transformations, Null spaces and Ranges

Definition. Let V and W be vector spaces. We call a function (or a map) $T : V \to W$ a linear transformation from V to W if for all $x, y \in V$, $c \in \mathbb{F}$ we have

(i) $T(x + y) = T(x) \oplus T(y)$

(ii) $T(cx) = c \odot T(x)$

Fact. Let $T : V \to W$ be a linear transformation. Then $T(\mathbb{O}) = \mathbb{O}$.

Proof. Let $x \in V$. Then $T(\mathbb{O}) = T(x - x) = T(x) - T(x) = \mathbb{O}$. □

Fact. Let $T : V \to W$ be a function. Then T is linear iff for $x_1, \ldots, x_n \in V$ and $c_1, \ldots, c_n \in \mathbb{F}$

$$T\Big(\sum_{i=1}^{n} a_i x_i\Big) = \sum_{i=1}^{n} a_i T(x_i)$$

Proof. The proof follows from the definition by induction. □

Definition. Let V be a vector space.

- $I_v : V \to V$ where $I_V(x) = x$ is called the identity transformation.
- $T_0 : V \to W$ where $T_0(x) = \mathbb{O}_W$ is called the zero transformation.

It is clear that both of these transformations are linear. We often write I instead of I_V.

Definition. Let V and W be vector spaces (over the same field $\mathbb{F}$) and $T : V \to W$ be linear. We define

- $N(T) \coloneqq \{v \in V : T(v) = \mathbb{O}\}$ called the null-space or kernel.
- $R(T) \coloneqq \{T(v) : v \in V\}$ called the column-space or range.

Theorem 2.1. Let V, W be vector spaces and let $T : V \to W$ be linear. Then $N(T)$ is a subspace of V and $R(T)$ is s a subspace of W.

Proof. Since $T(0) = 0$, $0 \in N(T)$. Now let v and w in $N(T)$. Then $T(v+w) = T(v)+T(w) = 0 + 0 = 0$. Similarly $T(cv) = cT(v) = c0 = 0$. Hence $v + w \in N(T)$ and $cv \in N(T)$, so that $N(T)$ is a subspace of V.
Now because $T(0) = 0$, $0 \in R(T)$. Then let $T(v)$ and $T(W)$ in $R(T)$. We have $T(v)+T(w) = T(v + w)$ and $cT(v) = T(cv)$. Hence $T(v + w) \in R(T)$ and $cT(v) \in R(T)$, so that $R(T)$ is a subspace of W. □

Example. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a function defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ where $a_1, a_2, a_3 \in \mathbb{R}$.
Then $N(T) = \text{span}\{(1, 1, 0)\}$ and $R(T) = \text{span}\{(1, 0), (0, 1)\} = \mathbb{R}^2$.

Theorem 2.3. Let V and W be vector spaces and $T : V \to W$ be linear. If V is finite dimension, then

$$\text{nullity}(T) + \text{rank}(T) = \dim V$$

where $\text{nullity}(T) := \dim N(T)$ and $\text{rank}(T) := \dim R(T)$.

Proof. Proof in textbook. □

Theorem 2.4. Let V and W be vector spaces and $T : V \to W$ be linear. Then T is one-to-one (injective) iff $N(T) = \{0\}$.

Proof. ($\Rightarrow$). Suppose T is one-to-one. For any $x \in N(T)$, if $T(x) = 0 = T(0)$, then $x = 0$. Hence $N(T) = \{0\}$.
($\Leftarrow$). Suppose $N(T) = \{0\}$. For any $x, y \in V$, if $T(x) = T(y)$ then $T(x-y) = T(x)-T(y) = 0 = T(0)$. Hence $x - y = 0$ and so $x = y$. That is T is one-to-one. □

Theorem 2.5. Let V and W be vector spaces of equal finite dimension and let $T : V \to W$ be linear. Then the following are equivalent:

(i) T is one-to-one (injective)

(ii) T is onto (surjective)

(iii) $\text{rank}(T) = \dim(V)$

Proof. Recall $\text{nullity}(T) + \text{rank}(T) = \dim(V)$.
(i $\Leftrightarrow$ iii) T is one-to-one $\Leftrightarrow$ $\text{nullity}(T) = 0 \Leftrightarrow \text{rank}(T) = \dim(V)$.
(ii $\Leftrightarrow$ iii) T is onto $\Leftrightarrow R(T) = W \Leftrightarrow \text{rank}(T) = \dim(V)$. □

Theorem 2.6. Let V and W be vector spaces over $\mathbb{F}$ and suppose that $\{v_1, \ldots, v_n\}$ is a basis for V. For any n vectors $w_1, \ldots, w_n \in W$, there exists exactly one linear transformation $T : V \to W$ such that $T(v_1) = w_1, \ldots, T(v_n) = w_n$.

Proof. Let $x \in V$. Then

$$x = a_1 v_1 + \cdots + a_n v_n$$

for $a_1, \ldots, a_n \in \mathbb{F}$ which are determined uniquely by x.
Define a map $T : V \to W$ by

$$T(x) = T(a_1 v_1 + \cdots + a_n v_n) \coloneqq a_1 w_1 + \cdots + a_n w_n$$

It is easy to check T is linear and $T(v_i) = w_i$ for $i = 1, \ldots, n$.
Now let us prove the uniqueness.
Suppose T and S both are linear transformation having the above property, then for all $x = a_1 v_1 + \cdots + a_n v_n \in V$, we have

$$T(x) = T\Big(\sum_{i=1}^{n} a_i v_i\Big) = \sum_{i=1}^{n} a_i T(v_i) = \sum_{i=1}^{n} a_i w_i = \sum_{i=1}^{n} a_i S(v_i) = S(x)$$

Hence $S = T$. □

Remark. Let $T : V \to W$ be linear. If $\{v_1, \ldots, v_k\}$ is a linearly independent subset of V then the image need not be linearly independent. In fact if $a_1 T(v_1) + \cdots + a_k T(v_k) = 0$ for some scalars, then $T(a_1 v_1 + \cdots + a_k v_k) = 0$ but $a_1 v_1 + \cdots + a_k v_k = 0$ is not a guarantee.
If $\{v_1, \ldots, v_k\}$ is linearly dependent, then $\{T(v_1), \ldots, T(v_k)\}$ is linearly dependent.

2.2 The Matrix Representation of a Linear Transformation

Definition. Let V be a vector space with finite dimension. An ordered basis is a basis for V endowed with a specific order, i.e. and ordered basis for V is a finite sequence of linearly independent vectors that span V.
For $\mathbb{F}^n$, we call $\{e_1, \ldots, e_n\}$ the standard ordered basis for $\mathbb{F}^n$.
For $P_n(\mathbb{F})$, we call $\{1, x, x_2, \ldots, x^n\}$ the standard ordered basis for $P_n(\mathbb{F})$.

Definition. Suppose $\beta = \{v_1, \ldots, v_n\}$ is an ordered basis for a finite dimensional vector space V. For any $x \in V$, let $a_1, \ldots, a_n$ be the unique scalars such that $x = a_1 v_1 + \cdots + a_n v_n$. We define the coordinate vector of x relative to β, denoted $[x]_\beta$, by

$$[x]_\beta = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Example. If $x = a_1v_1 + \cdots + a_nv_n$, $\beta = \{v_1, \ldots, v_n\}$ and $\gamma = \{v_2, v_3, v_1, v_4, \ldots, v_n\}$ for a n dimensional vector space V, then

$$[x]_\beta = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ \vdots \\ a_n \end{bmatrix}, \ [x]_\gamma = \begin{bmatrix} a_2 \\ a_3 \\ a_1 \\ a_4 \\ \vdots \\ a_n \end{bmatrix}$$

Definition. Suppose that V and W are finite dimension vector spaces with ordered bases $\beta = \{v_1, \ldots, v_n\}$ and $\gamma = \{w_1, \ldots, w_m\}$ respectively.
Let $T : V \to W$ be linear, then for each $j = 1, \ldots, n$ we have $T(v_j) = \sum_{i=1}^m a_{ij}w_i$ and

$$[T(v_j)]_\gamma = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

We call the $m \times n$ matrix

$$\begin{aligned} A &= \begin{bmatrix} a_{1,1} & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \ldots & a_{m,n} \end{bmatrix} \\ &= [T(v_1), \ldots, T(v_n)] \end{aligned}$$

the matrix representation of T in the ordered bases β and γ and we write $A = [T]_\beta^\gamma$. A is sometimes called the matrix of T in the ordered bases β and
In the case where $V = W$ and $\beta = \gamma$ (as ordered bases) then we write $A = [T]_\beta$. γ.

Example. Let $T : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ be the linear transformation defined by $T(f(x)) \coloneqq f'(x)$. Let β and γ be the standard ordered bases for $P_3(\mathbb{R})$ and $P_2(\mathbb{R})$ respectively. Then

$$\begin{aligned} A = [T]_\beta^\gamma &= ([T(1)]_\gamma, [T(x)]_\gamma, [T(x^2)]_\gamma, [T(x^3)]_\gamma) \\ &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \end{aligned}$$

Definition. Let $T, U : V \to W$ be two functions where V and W are vector spaces over the same field $\mathbb{F}$ and let $a \in \mathbb{F}$. Then we define

$$T + U : V \to W \text{ and } aT : V \to W$$

as follows:

(1) $(T+U)(x) \coloneqq T(x) = U(x), \quad \forall x \in V$

(2) $(aT)(x) \coloneqq aT(x)$

Definition. Let V and W be two vector spaces. We denote the vector space of all linear transformations from V into W by $\mathcal{L}(V, W)$.

Theorem 2.8. Let V and W be finite dimension vector spaces with ordered bases β and γ respectively and let $T, U \in \mathcal{L}(V, W)$. Then

(a) $[T+U]_\beta^\gamma = [T]_\beta^\gamma + [U]_\beta^\gamma$

(b) $[cT]_\beta^\gamma = c[T]_\beta^\gamma$

2.3 Composition of Linear Transformations and Matrix Multiplication

Theorem 2.9. Let V, W, Z be vector spaces over $\mathbb{F}$ and let $T : V \to W$, $U : W \to Z$ be linear. Then $U \circ T : V \to Z$ is linear and we define $UT \coloneqq U \circ T$.

Proof. Let $x, y \in V$ and $a \in \mathbb{F}$. Then

$$UT(ax+y) = U(T(ax+y)) = U(aT(x)+T(y)) = aU(T(x))+U(T(y)) = a(UT)(x)+(UT)(y)$$

□

Theorem 2.10. Let V be a vector spaces. Let $T, U_1, U_2 \in \mathcal{L}(V)$. Then

(a) $T(U_1 + U_2) = TU1 + TU_2$ and $(U1 + U_2)T = U_1T + U_2T$

(b) $T(U_1U_2) = (TU_1)U_2$

(c) $TI = IT = T$

(d) $a(U_1U_2) = (aU_1)U_2 = U_1(aU_2)$ for all scalars a.

Theorem 2.11. Let V, W and Z be finite dimensional vector spaces with ordered bases α, β, γ respectively. Let $T : V \to W$ and $U : W \to Z$ be linear. Then

$$[UT]_\alpha^\gamma = [U]_\beta^\gamma [T]_\alpha^\beta$$

is linear.

Corollary 2.11.1. Let V be a finite dimension vector space with ordered basis β. Let $T, U \in \mathcal{L}(V)$. Then $[UT]_\beta = [U]_\beta [T]_\beta$.

Theorem 2.14. Let V and W be finite dimensional vector spaces having ordered bases β and γ respectively. Let $T : V \to W$ be linear. Then for each $u \in V$, we have

$$[T(u)]_\gamma = [T]_\beta^\gamma [u]_\beta$$

Proof. Let $u \in V$ and define the following linear transformations $f : \mathbb{F} \to V$ by $f(a) = au$ and $g : \mathbb{F} \to W$ by $g(a) = aT(u)$. Let $\alpha = \{1\}$ be the standard basis for $\mathbb{F}$.
Notice $g = Tf$. Then by Theorem 2.11

$$[T(u)]_\gamma = [g(1)]_\gamma = [g]_\alpha^\gamma = [Tf]_\alpha^\gamma = [T]_\beta^\gamma [f]_\alpha^\beta = [T]_\beta^\gamma [f(1)]_\beta = [T]_\beta^\gamma [u]_\beta$$

□

Definition. Let A be a $m \times n$ matrix with entries from $\mathbb{F}$. We define the mapping $L_A : \mathbb{F}^n \to \mathbb{F}^m$ be $L_A(x) = Ax$ for all $x \in \mathbb{F}^n$.
L_A is called the left-multiplication transformation.

Theorem 2.15. Let A be a $m \times n$ matrix with entries from a field $\mathbb{F}$. Then the left-multiplication $L_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear. Furthermore if B is any other $m \times n$ matrix and β and γ are the standard bases for $\mathbb{F}^n$ and $\mathbb{F}^m$ respectively then the following hold:

(a) $[L_A]_\beta^\gamma = A$

(b) $L_A = L_B$ iff $A = B$.

(c) $L_{A+B} = L_A + L_B$ and $L_{aA} = aL_A$

(d) If $T : \mathbb{F}^n \to \mathbb{F}^m$ is a linear transformation, then there exists a unique $m \times n$ matrix C such that $T = L_C$

(e) If E is a $n \times p$ matrix, then $L_{AE} = L_A L_E$

(f) If $m = n$, then $L_{I_n} = I_{\mathbb{F}_n}$

2.4 Invertability and Isomorphisms

Definition. Let V and W be vector spaces, and let $T : V \to W$ be linear. A function $U : W \to V$ is said to be an inverse of T if $TU = I_W$ and $UT = I_V$. The inverse of T (if T is invertible) is unique and we use T^{-1} to denote this inverse.

Remark.

$$(TU)^{-1} = U^{-1}T^{-1} \quad \text{and} \quad (T^{-1})^{-1} = T$$

Remark. Let $T : V \to W$ be a linear transformation where V and W are finite dimensional vector spaces of same dimension. Then

$$T \text{ is invertible} \Leftrightarrow \operatorname{rank}(T) = \dim(V)$$

Definition. Let A be a $n \times n$ matrix. Then A is invertible[1] if there another $n \times n$ matrix B such that

$$AB = I_n = BA$$

[1] A matrix is invertible iff its determinant is non-zero. Furthermore invertible matrices are bijective.

Fact. Let T be an invertible linear transformation from $V \to W$. Then V is finite dimensional iff W is finite dimensional. In this case, $\dim(V) = \dim(W)$.

Proof. Suppose V is finite dimension, let β be a basis for V and $T(\beta) = \{T(x_i)|x_i \in \beta\}$. Then $R(T) = \text{span}(T(\beta))$. Note that $R(T) \subseteq W$. In fact it's enough to prove that $W \subseteq R(T)$. For any $y \in W$, we have $TU = I_W$ for the inverse U of T and $y = I_W(y) = T(U(y)) \in R(T)$. So $W \subseteq R(T)$ i.e. $W = R(T)$. Hence W is finite dimensional.
Conversely, if W is finite dimensional, then using the fact $T^{-1} : W \to V$ is linear and what we have proven, V is finite dimensional. □

Theorem 2.18. Let V and W be finite dimension vector spaces with ordered bases β and γ respectively. Let $T : V \to W$ be linear. Then T is invertible iff $[T]_\beta^\gamma$ is invertible. Moreover,

$$[T^{-1}]_\gamma^\beta = ([T]_\beta^\gamma)^{-1}$$

Proof. Suppose that T is invertible. Then $\dim(V) = \dim(W)$ by the previous lemma. Let $n = \dim(V) = \dim(W)$. Then $[T]_\beta^\gamma$ is a $n \times n$ matrix. Let $T^{-1} : W \to V$ be the inverse of T. Then $TT^{-1} = I_W$ and $T^{-1}T = I_V$. Thus

$$I_n = [I_w]_\gamma^\gamma = [TT^{-1}]_\gamma^\gamma = [T]_\beta^\gamma [T^{-1}]_\gamma^\beta \tag{1}$$

Similarly,

$$I_n = [T^{-1}]_\gamma^\beta [T]_\beta^\gamma \tag{2}$$

By (1) and (2), $([T]_\beta^\gamma)^{-1} = [T^{-1}]_\gamma^\beta$.
Now suppose that $A = [T]_\beta^\gamma$ is invertible. Then there exists a $n \times n$ matrix B such that $AB = BA = I_n$. Then by Theorem 2.6, there exists $U \in \mathcal{L}(W, V)$ such that

$$U(w_j) = \sum_{i=1}^{n} B_{ij} v_i \quad \text{for } j = 1, 2, \ldots, n,$$

where $\gamma = \{w_1, \ldots, w_n\}$ and $\beta = \{v_1, \ldots, v_n\}$. It follows that $[U]_\gamma^\beta = B$. To show that $U = T^{-1}$, observe that

$$[UT]_\beta = [U]_\gamma^\beta [T]_\beta^\gamma = BA = I_n = [I_V]_\beta$$

by Theorem 2.11. So $UT = I_V$ and similarly $TU = I_W$. □

Definition. Let V and W be vector spaces. We say V is isomorphic to W if there exists a linear transformation $T : V \to W$ that is invertible. Such an invertible linear transformation from $V \to W$ is called an isomorphism from V to W.

Example. Consider $V = P_3(\mathbb{R})$ and $W = M_{2\times 2}(\mathbb{R})$.
Define $T : P_3(\mathbb{R}) \to M_{2\times 2}(\mathbb{R})$ by

$$T(f) = \begin{bmatrix} f(1) & f(2) \\ f(3) & f(4) \end{bmatrix}$$

for all $f \in P_3(\mathbb{R})$. Then

$$\begin{aligned} T(f+g) &= \begin{bmatrix} (f+g)(1) & (f+g)(2) \\ (f+g)(3) & (f+g)(4) \end{bmatrix} \\ &= \begin{bmatrix} f(1)+g(1) & f(2)+g(2) \\ f(3)+g(3) & f(4)+g(4) \end{bmatrix} \\ &= \begin{bmatrix} f(1) & f(2) \\ f(3) & f(4) \end{bmatrix} + \begin{bmatrix} g(1) & g(2) \\ g(3) & g(4) \end{bmatrix} = Tf + Tg \end{aligned}$$

and

$$T(cf) = cTf$$

So T is linear. Now since $\dim(P_3(\mathbb{R})) = 4 = \dim(M_{2\times 2}(\mathbb{R})$, it's enough to prove that T is injective to get the invertability of T. If $f \in N(T)$ then

$$Tf = O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} f(1) & f(2) \\ f(3) & f(4) \end{bmatrix}$$

and so $f(1) = f(2) = f(3) = f(4) = 0$ which implies $f = 0$.

Fact. Let V and W be finite dimensional vector spaces of dimension n and m respectively. Then $\mathcal{L}(V, W)$ is finite dimensional and of dimension mn.

Proof. Define $\Phi : \mathcal{L}(V, W) \to M_{m\times n}(\mathbb{F})$ by

$$\Phi(t) := [T]_\beta^\gamma$$

for all $T \in \mathcal{L}(V, W)$, where β is a basis for V and γ is a basis for W. Then one can check Φ is an isomorphism. Hence $\dim(\mathcal{L}(V, W) = \dim(M_{m\times n}(\mathbb{F}) = mn$. □

Definition. Let β be an ordered basis for a n-dimensional vector space V. The standard representation of V with respect to β is the function $\phi_\beta : V \to \mathbb{F}^n$ defined by $\phi_\beta(x) = [x]_\beta$ for all $x \in V$.

Theorem 2.21. For any finite dimensional vector space V with an ordered basis β, ϕ_β is an isomorphism.

Remark. Let V and W be two vector spaces with $\dim(V) = n$ and $\dim(W) = m$. Let $T : V \to W$ be linear. Then the following diagram is commutative:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\phi_\beta"'] & W \arrow[d, "\phi_\gamma"] \\ \mathbb{F}^n \arrow[r, "L_A"'] & \mathbb{F}^m \end{tikzcd}$$

where $A = [T]_\beta^\gamma$, β and γ are a bases for V and W respectively. We can see that $\phi_\gamma T = L_A \phi_\beta$.

Example. Recall the linear transformation $T : P_3(\mathbb{R}) \to P_2(\mathbb{R}$ defined by $T(f(x)) = f'(x)$. Let β and γ be the standard bases for $P_3(\mathbb{R})$ and $P_2(\mathbb{R})$ respectively. Then

$$A = [T]_\beta^\gamma = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

and $L_A\phi_\beta = \phi_\gamma T$. In fact if $a + bx + cx^2 + dx^3 \in P_3(\mathbb{R})$ with $a, b, c, d \in \mathbb{F}$, then

$$(L_A\phi_\beta)(f) = L_A\phi_\beta(f) = L_A[f]_\beta = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} b \\ 2c \\ 3d \end{bmatrix}$$

and

$$(\phi_\gamma T)(f) = \phi_\gamma T(f) = \phi_\gamma(f') = \phi_\gamma(b + 2cx + 3dx^3) = \begin{bmatrix} b \\ 2c \\ 3d \end{bmatrix}$$

Hence $L_A\phi_\beta = \phi_\gamma T$.

2.5 The Change of Coordinate Matrix

Theorem 2.22. Let β and β' be two ordered bases for a finite dimension vector space V and let $[I_V]_{\beta'}^\beta = Q$. Then

(a) Q is invertible.

(b) For any $v \in V$, $Q[v]_{\beta'} = [v]_\beta$

Proof.

(a) Since I_V is invertible, so is Q by Theorem 2.18.

(b) For any $v \in V$

$$[v]_\beta = [I_V(v)]_\beta = [I_V]_{\beta'}^\beta [v]_{\beta'} = Q[v]_{\beta'}$$

by Theorem 2.14.

□

Example. Let $\beta = \{(1, 1), (1, -1)\}$, $\beta' = \{(2, 4), (3, -1)\}$.
Then

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$
$$\begin{bmatrix} 3 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Giving the following matrix for Q:

$$Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$$

Then for $u = (2, 4)$, $[u]_\beta = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $[u]_{\beta'} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Furthermore

$$Q[v]_{\beta'} = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Definition. The matrix Q defined in Theorem 2.22 is called the change of coordinate matrix. We say that Q changes β'-coordinates $[v]_{\beta'}$ into β-coordinates $[v]_\beta$.

Theorem 2.23. Let T be a linear transformation from V to V where V is a finite dimensional vector space. Let β and β' be two ordered bases for V. Suppose $Q = [I_V]_\beta$. Then

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q$$

Proof. Let I_V be the identity transformation from V to V. Then $TI_V = T = I_V T$ and

$$Q[T]_{\beta'} = [I_V]_{\beta'}^{\beta}[T]_{\beta'}^{\beta'} = [IT]_{\beta'}^{\beta} = [TI]_{\beta'}^{\beta} = [T]_{\beta}^{\beta}[I_V]_{\beta'}^{\beta} = [T]_\beta Q$$

So

$$[T]_{\beta'} = Q^{-1}Q[T]_\beta = Q^{-1}[T]_\beta Q$$

□

Note. A general strategy to solve this is find the linear composition of each vector in β' in terms of vectors in β, giving the corresponding column of the change of coordinate matrix.

2.6 Dual Spaces

Definition. Let V be a vector space over a field $\mathbb{F}$. A linear transformation $f : V \to \mathbb{F}$ is a called a linear functional on V.

Example. Let $V = M_{n\times n}(\mathbb{F})$ and define $f : V \to \mathbb{F}$ by $f(A) = \operatorname{tr}(A)$ for $A \in M_{n\times n}(\mathbb{F})$ where $\operatorname{tr}(A) = \sum_{i=0}^{n} a_{i,i}$ and $A = (a_{i,i})_{1\le i,j\le n}$. Then f is a linear functional.

Example. Let V be a finite dimensional vector space and $\beta = \{x_1, \ldots, x_n\}$ be an ordered basis for V. For each $i = 1, \ldots, n$ we define $f_i : V \to \mathbb{F}$ by $f_i(x_i) = a_i$ where

$$[x]_\beta = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

Then f_i is a linear functional on V and is called the i-th coordinate function with respect to the basis β.

Definition. For a vector space V of $\mathbb{F}$, we define the dual space of V to be the vector space $\mathcal{L}(V, \mathbb{F})$ denoted by V^* and $\dim(V) = \dim(V^*)$.

Theorem 2.24. Suppose V is a finite dimensional vector space with the ordered basis $\beta = \{x_1, \ldots, x_n\}$. Let f_i be the i-th coordinate function with respect to β and let $\beta^* = \{f_1, f_2, \ldots, f_n\}$. Then β^* is a basis for V^* and any $f \in V^*$ can be expressed by $f = \sum_{i=1}^n f(x_i) f_i$.

Proof. It is enough to prove $V^* = \operatorname{span}(\beta^*)$. Let $f \in V^*$. We are going to prove that f be written in the form given above.
For all $y \in V$, we have $y = a_1 x1 + \cdots + a_n x_n$ for some scalars $a_1, \ldots, a_n$. Then

$$f(y) = f\Big(\sum_{i=1}^n a_i x_i\Big) = \sum_{j=1}^n a_j f(x_j) \tag{1}$$

and

$$\begin{aligned}
\Big(\sum_{i=1}^n f(x_i) f_i\Big)(y) &= \Big(\sum_{i=1}^n f(x_i))\Big) f_i(y) \\
&= \sum_{i=1}^n f(x_i) f_i\Big(\sum_{j=1}^n a_j x_j\Big) \\
&= \sum_{i=1}^n f(x_i) \sum_{j=1}^n a_j f_i(x_j) \\
&= \sum_{j=1}^n \sum_{i=1}^n f(x_j) a_j f_i(x_j) \\
&= \sum_{j=1}^n a_j \Big(\sum_{i=1}^n f(x_i) f_i\Big)(x_j) \qquad (2)
\end{aligned}$$

We will now show $f(x_j) = \sum_{i=1}^n (f(x_i) f_i(x_j)$

$$\begin{aligned}
f(x_j) &= \sum_{i=1}^n (f(x_i) f_i(x_j) \\
&= \sum_{i=1}^n (f(x_i) f_i)(x_j) \\
&= \sum_{i=1}^n f(x_i) f_i(x_j) \\
&= \sum_{i=1}^n f(x_j) \delta_{i,j} \\
&= f(x_j) \delta_{j,j} = f(x_j) \qquad (3)
\end{aligned}$$

By (1), (2) and (3) the Theorem is true. □

Definition. Using the notation of Theorem 2.24, we call the ordered basis $\beta^* = \{f_1, \ldots, f_n\}$ of V^*, that satisfies $f_i(x_j) = \delta_{i,j} (1 \leq i, j \leq n)$, the dual basis of β.

Example. Let $\beta = \{(2,1),(3,1)\}$. Find the dual basis β^* of β.
Let β be the ordered basis for $\mathbb{R}^2$ as above. Then

$$\begin{aligned} f_1(2,1) = 1 = f_1(2e_1 + e_2) = 2f_1(e_1) = f_1(e_2) \\ f_1(3,1) = 0 = f_1(3e_1 + e_2) = 3f_1(e_1) = f_1(e_2) \end{aligned}$$

Solving this system gives $f_1(e_1) = -1, f_1(e_2) = 3$. Hence $f_1(x,y) = -x + 3y$ for $(x,y) \in \mathbb{R}^2$.
Similarly, $f_2(x,y) = x - 2y$ for $(x,y) \in \mathbb{R}^2$.

Example. Let $V = P_1(\mathbb{R})$ and for $p(x) \in V$, we define $f_1, f_2 \in V^*$ by

$$f_1(p(x)) = \int_0^1 p(t)\mathrm{d}t \quad \text{and} \quad f_2(p(x) = \int_0^2 p(t)\mathrm{d}t$$

Prove that f_1, f_2 is a basis for V^* and find a basis for which it is the dual basis.

Proof. It's enough to prove that f_1 and f_2 are linearly independent. Suppose $a_1 f_1 + a_2 f_2 = 0$ for some scalars a_1, a_2. Then

$$\begin{aligned} 0 = (a_1 f_1 + a_2 f_2)(1) = a_1 f_1(1) + a_2 f_2(1) \\ 0 = (a_1 f_1 + a_2 f_2)(x) = a_1 f_1(x) + a_2 f_2(x) \end{aligned} \tag{*}$$

So

$$\begin{aligned} f_1(1) = \int_0^1 1\mathrm{d}t = t\big|_0^1 = 1 \quad f_1(x) = \int_0^1 t\mathrm{d}t = \frac{t^2}{2}\Big|_0^1 = \frac{1}{2} \\ f_2(1) = \int_0^2 1\mathrm{d}t = t\big|_0^2 = 2 \quad f_2(x) = \int_0^1 2\mathrm{d}t = \frac{t^2}{2}\Big|_0^2 = 2 \end{aligned}$$

and (*) becomes

$$\begin{aligned} a_1 + 2a_2 = 0 \\ \frac{1}{2}a_1 + 2a_2 = 0 \end{aligned}$$

Since this system only has the trivial solution, $a_1 = a_2 = 0$.
Hence $\{f_1, f_2\}$ is a basis for V^*.
Now let $\beta = \{c_0 + c_1 x, d_0 + d_1 x\}$ be a basis for V, where c_0, c_1, d_0 and d_1 are scalars.
If $\{f_1, f_2\}$ is a dual basis for β, then

$$\begin{aligned} f_1(c_0 + c_1 x) = 1 \quad f_2(c_0 + c_1 x) = 1 \\ f_1(d_0 + d_1 x) = 0 \quad f_2(d_0 + d_1 x) = 0 \end{aligned}$$

Note that

$$f_1(c_0 + c_1 x) = \int_0^1 c_0 + c_1 x) = \left(c_0 t + \frac{c_1 t^2}{2}\right)\Big|_0^1 = c_0 + \frac{c_1}{2} = 1 \tag{1}$$

Similarly,

$$2c_0 + 2c_1 = 0 \tag{2}$$

$$d_0 + \frac{1}{2} d_1 = 0 \tag{3}$$

$$2d_0 + 2d_1 = 1 \tag{4}$$

(1) and (2) give $c_0 = 2, c_1 = -2$.
(3) and (4) give $d_0 = \frac{-1}{2}, d_1 = 1$.
It follows that $\{f_1, f_2\}$ is the dual basis of $\beta = \{2 - 2x, \frac{-1}{2} + x\}$. $\square$

Chapter 4

Determinants

4.1 Some Reminders about Determinants

Let $A_{m\times n}$ with entries from $\mathbb{F}$. Then

$$\begin{aligned}\det A &= \sum_{j=1}^{n}(-1)^{i+j}A_{i,j}\det\widetilde{A_{i,j}} \quad \text{for } i=1,\dots,m\\ &= \sum_{i=1}^{m}(-1)^{i+j}A_{i,j}\det\widetilde{A_{i,j}} \quad \text{for } j=1,\dots,n\end{aligned}$$

where $\widetilde{A_{i,j}}$ is a $(n-1)\times(m-1)$ matrix defined by deleting the i-th row and j-th column of the matrix A.

Let us recall some useful properties of the determinant.

- $\det A = \det A^T$.
- $\det AB = \det A \det B$.
- A is invertible $\Leftrightarrow \det A \neq 0$.
- Interchanging a row or a column flips the sign of the determinant.
- Scaling a row or a column by a constant scales the determinant by the same constant.
- Adding a multiple of one row to another does not change the determinant.
- $\det\left[\begin{array}{c|c} A_{n\times n} & C_{n\times m} \\ \hline 0_{m\times n} & B_{m\times m}\end{array}\right] = \det A \det B$

Chapter 5

Diagonalization

5.1 Eigenvalues and Eigenvectors

Definition. A linear transformation (or operator) T on a finite dimensional vector space V is called diagnalizable if there is an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix. A square matrix A is diagonalizable if L_A is diagonalizable.

Definition. Let T be a linear operator on a vector space V¿ A non-zero vector $v \in V$ is called an eigenvector of T if $T(v) = \lambda v$ for some scalar λ. The scalar λ is called the eigenvalue corresponding to the eigenvector v.
Let $A \in M_{n\times n}(\mathbb{F})$. A non-zero vector $v \in \mathbb{F}^n$ is an eigenvector of A if $Av = \lambda v$ for some scalar λ. This scalar is called the eigenvalue corresponding to the eigenvector v.

Theorem 5.1. A linear operator T on a finite dimensional vector space is diagonalizable if and only if there exists an ordered basis β consisting of eigenvectors of T. Furthermore, if T is diagonalizable, $\beta = \{v_1, \ldots, v_n\}$ is an ordered basis of eigenvectors of T and $D = [T]_\beta$, then D is a diagonal matrix and $D_{j,j}$ is the eigenvalue corresponding to v_j.

Proof. ($\Rightarrow$) Suppose T is diagonalizable. Then there exists an ordered basis β such that

$$[T]_\beta = \begin{bmatrix} D_{1,1} & & & \\ & D_{2,2} & & \\ & & \ddots & \\ & & & D_{n,n} \end{bmatrix} = D.$$

However

$$[T]_\beta = (T[(v_1)]_\beta, \ldots, T[(v_n)]_\beta) \quad \text{and} \quad [T]_\beta[v_1]_\beta = \begin{bmatrix} D_{1,1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}_\beta = D_{1,1}v_1.$$

More generally

$$[T(v_j)]_\beta = D_{j,j}v_{j,j} \qquad (1 \le j \le n).$$

So β is an ordered basis for V consisting of eigenvectors.
($\Leftarrow$) Exercise. $\square$

Theorem 5.2. Let $A \in M_{n\times n}(\mathbb{F})$. Then a scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof. A scalar λ is an eigenvalue of A if and only if there exists a non-zero vector $v \in \mathbb{F}^n$ such that $Av = \lambda v$, that is, $(A-\lambda I_n)v = 0$. By Theorem 2.5 this is true if and only if $A-\lambda I_n$ is invertible. However this is equivalent to the statement that $\det(A - \lambda I_n) = 0$. □

Definition. Let $A \in M_{n\times n}(\mathbb{F})$. The polynomial $f(t) = \det(A - tI_n)$ is called the characteristic polynomial of A.

Definition. Let T be a linear operator on a n-dimensional vector space V with ordered basis β. We define the characteristic polynomial $f(t)$ of T to be the characteristic polynomial of $[T]_\beta$.

Remark. The characteristic polynomial of a linear operator does not depend on the choice of the basis for V.

Example. Let T be the linear operator of $P_2(\mathbb{R})$ defined by

$$T(f(x)) = f(x) + (x+1)f'(x+1) \qquad (f(x) \in P_2(\mathbb{R})).$$

Let β be the standard ordered basis of $P_2(\mathbb{R})$. Then

$$A = [T]_\beta = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

The characteristic polynomial $f(t)$ is given by

$$f(t) = \det(A - tI_3) = \begin{vmatrix} 1-t & 1 & 0 \\ 0 & 2-t & 2 \\ 0 & 0 & 3-t \end{vmatrix} = (t-1)(t-2)(t-3).$$

Theorem 5.3. Let $A \in M_{n\times n}(\mathbb{F})$.

(a) The characteristic polynomial of A is a polynomial of degree n with leading coefficient $(-1)^n$.

(b) A has at most n distinct eigenvalues.

Proof. (a) Let us proceed by induction.

Base case: Let $n = 2$ and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Then

$$\det(A - \lambda I_2) = (a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + ad - bc$$

So the characteristic polynomial of A is a polynomial of degree 2 with leading coefficient 1 as required.

Inductive step: Let $n \geq 2$. Suppose suppose that if $B \in M_{n\times n}(\mathbb{F})$ then the characteristic polynomial of A of degree n with leading coefficient $(-1)^n$.

Let $A \in M_{(n+1)\times(n+1)}$.

Then

$$\det(A - \lambda I_{n+1}) = (a_{1,1} - \lambda)\det \widetilde{A_{1,1}} - (a_{1,2})\det \widetilde{A_{1,2}} + \cdots + (-1)^{1+(n+1)}\det \widetilde{A_{1,(n+1)}} \quad (1)$$

where $\widetilde{A_{i,j}}$ is a $(n-1)\times(n-1)$ matrix defined by deleting the i-th row and j-th column of $(A-\lambda I_{n+1})$. Notice all the determinants of sub-matrices is a degree n polynomial. Hence all summands except the first one are polynomials of degree n, and the first summand is

$$(a_{1,1} - \lambda)p(\lambda)$$

Where $p(\lambda) = (-\lambda)^n + c_{n-1}\lambda^{n-1} + \dots$.

So

$$(a_{1,1} - \lambda)p(\lambda) = (-\lambda)^{n+1} + c_{n-1}\lambda^n + \dots$$

Hence the first summand of (1) is a polynomial of degree $n+1$ with leading coefficient $(-1)^{n+1}$, and so $\det(A - \lambda I_{n+1}$ is a polynomial of degree $n+1$ with leading coefficient $(-1)^{n+1}$ as required.

□

Theorem 5.4. Let T be a linear operator on a finite dimensional vector space V and let λ be an eigenvalue of T.
A vector $v \in V$ is an eigenvector of T if and only if $v \neq 0$ and $v \in N(T - \lambda I)$.

To find the eigenvectors of a linear operator T on a finite dimensional vector space V, select an ordered basis β for V and $A = [T]_\beta$. Then the following diagram is commuatative:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\phi_\beta"'] & W \arrow[d, "\phi_\gamma"] \\ \mathbb{F}^n \arrow[r, "L_A"'] & \mathbb{F}^m \end{tikzcd}$$

Fact. $v \in V$ is an eigenvector of T with corresponding eigenvalue λ if and only if $\phi_\beta(v)$ is an eigenvector of A with the corresponding eigenvalue λ.

Proof. ($\Rightarrow$) Suppose $T(v) = \lambda v$. Then

$$A\phi_\beta(v) = L_A\phi_\beta(v) = \phi_\beta T(v) = \phi_\beta(\lambda v).$$

It follows that

$$A\phi_\beta(v) = \lambda\phi_\beta(v).$$

($\Leftarrow$) Suppose $A\phi_\beta(v) = \lambda\phi_\beta(v)$. Then

$$\phi_\beta(\lambda v) = \lambda\phi_\beta(v) = A\phi_\beta(v) = L_A\phi_\beta(v) = \phi_\beta T(v).$$

It follows that

$$\phi_\beta(\lambda v) = \phi_\beta T(v).$$

Hence

$$T(v) = \phi_\beta^{-1}\phi_\beta T(v) = \phi_\beta^{-1}\phi_\beta(\lambda v) = \lambda v.$$

□

5.2 Diagonalizability

Theorem 5.5. Let T be a linear operator on a finite dimensional vector space V and let $\lambda_1, \lambda_2, \ldots, \lambda_k$ be distinct eigenvalues of T. If $v_1, v_2, \ldots, v_k$ are the eigenvectors of T such that λ_i corresponds to v_i for $1 \leq i \leq k$. Then $\{v_1, \ldots, v_k\}$ is linearly independent.

Proof. Suppose these k vectors are not linearly independent. Then

$$a_1v_1 + \cdots + a_kv_k = 0$$

for the scalars $a_1, \ldots, a_k$ which are not all 0. After reordering $v_1, \ldots, v_k$, we can assume

$$a_1v_1 + \cdots + a_pv_p = 0$$

with $a_1 \cdot \cdots \cdot a_p \neq 0$. Then

$$0 = T(a_1v_1 + \cdots + a_pv_p) = a_1T(v_1) + \cdots + a_pT(v_p) = a_1\lambda_1v_1 + \cdots + a_p\lambda_pv_p$$

which in turn implies

$$0 = T(a_1\lambda_1v_1 + \cdots + a_p\lambda_pv_p) = a_1\lambda_1^2v_1 + \cdots + a_p\lambda_p^2v_p.$$

In this way, we get the following system:

$$\begin{aligned} a_1v_1 + \cdots + a_pv_p &= 0 \\ a_1\lambda_1v_1 + \cdots + a_p\lambda_pv_p &= 0 \\ a_1\lambda_1^2v_1 + \cdots + a_p\lambda_p^2v_p &= 0 \\ &\vdots \\ a_1\lambda_1^{p-1}v_1 + \cdots + a_p\lambda_p^{p-1}v_p &= 0 \end{aligned}$$

Which is equivalent to

$$\begin{bmatrix} a_1 & a_2 & \ldots & a_p \\ a_1\lambda_1 & a_2\lambda_2 & \ldots & a_p\lambda_p \\ a_1\lambda_1^2 & a_2\lambda_2^2 & \ldots & a_p\lambda_p^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1\lambda_1^{p-1} & a_2\lambda_2^{p-1} & \ldots & a_p\lambda_p^{p-1} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_p \end{bmatrix} = 0.$$

Then

$$\det \begin{bmatrix} a_1 & a_2 & \dots & a_p \\ a_1\lambda_1 & a_2\lambda_2 & \dots & a_p\lambda_p \\ a_1\lambda_1^2 & a_2\lambda_2^2 & \dots & a_p\lambda_p^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1\lambda_1^{p-1} & a_2\lambda_2^{p-1} & \dots & a_p\lambda_p^{p-1} \end{bmatrix} = a_1a_2\dots a_p \det \begin{bmatrix} 1 & 1 & \dots & 1 \\ \lambda_1 & \lambda_2 & \dots & \lambda_p \\ \lambda_1^2 & \lambda_2^2 & \dots & \lambda_p^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{p-1} & \lambda_2^{p-1} & \dots & \lambda_p^{p-1} \end{bmatrix} \neq 0.$$

Hence $v_1 = v_2 = \dots = v_p = 0$ which is a contradiction. □

Corollary 5.5.1. Let T be a linear operator on an n-dimensional vector space V. If T has n distinct eigenvalues, then T is diagonalizable.

Definition. A polynomial $f(t) \in P(\mathbb{F})$ splits over $\mathbb{F}$ if there exist scalars $c, a_1, \dots, a_n \in \mathbb{F}$ such that

$$f(t) = c(t - a_1)(t - a_2)\dots(t - a_n).$$

Definition. Let λ be an eigenvalue of a linear operator or matrix with characteristic polynomial $f(t)$. The (algebraic) multiplicity of λ is the largest positive integer k for which $(t-\lambda)^k$ is a factor of $f(t)$.[1]

Example. Let

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 4 \end{bmatrix}$$

The characteristic polynomial of A is given by $f(t) = \det(A - tI_3) = (3-t)^2(4-t)$.
So $f(t)$ splits, the (algebraic) multiplicity of $\lambda_1 = 3$ is 2, and the multiplicity of $\lambda_2 = 4$ is 1.

Definition. Let T be a linear operator on a vector space V and let λ be an eigenvalue of T. Let

$$E_\lambda = \{x \in V | T(x) = \lambda x\} = N(T - \lambda I).$$

Then E_λ is called the eigenspace of T corresponding to the eigenvalue λ. $\dim(E_\lambda)$ is called the geometric multiplicity of λ.

Theorem 5.7. Let T be a linear operator on a finite dimensional vector space V and let λ be an eigenvalue of T having multiplicity m. Then

$$1 \leq \dim(E_\lambda) \leq m.$$

Proof. Let $\{v_1, \dots, v_p\}$ be a basis for E_λ. We extend $\{v_1, \dots, v_p\}$ to an ordered basis β for V

$$\beta = \{v_1, \dots, v_p, v_{p+1}, \dots, v_n\}$$

[1] Often times we will omit "algebraic" and call this the multiplicity of λ.

where $n = \dim(V)$. Since $T(v_i) = \lambda v_i$ for $1 \leq i \leq p$, then

$$A = [T]_\beta = [[T(v_1)]_\beta \dots [T(v_p)]_\beta [T(v_{p+1})]_\beta \dots [T(v_n)]_\beta]$$
$$= \left[\begin{array}{cccc|ccc} \lambda & 0 & \dots & 0 & * & \dots & * \\ 0 & \lambda & \dots & 0 & * & \dots & * \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda & * & \dots & * \\ \hline 0 & 0 & \dots & 0 & * & \dots & * \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & * & \dots & * \end{array}\right] = \left[\begin{array}{c|c} \lambda I_p & B \\ \hline 0 & C \end{array}\right]$$

Hence the characteristic polynomial is given by

$$\begin{aligned} f(t) = \det(A - tI_n) = \det([T]_\beta - tI_n) &= \det\left(\left[\begin{array}{c|c} \lambda I_p & B \\ \hline 0 & C \end{array}\right] - tI_n\right) \\ &= \det\left[\begin{array}{c|c} \lambda I_p - tI_p & B \\ \hline 0 & C - tI_{n-p} \end{array}\right] \\ &= \det(\lambda I_p - tI_p) \cdot \det(C - tI_{n-p}) \\ &= (\lambda - t)^p \det(C - tI_{(n-p)}). \end{aligned}$$

Hence $\dim(E_\lambda) = p \leq n$. □

Lemma. Let T be a linear operator and let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T. For each $i = 1, 2, \dots, k$, let $v_i \in E_{\lambda_i}$, where E_{λ_i} is the eigenspace corresponding to the eigenvalue λ_i. Then

$$v_1 + \dots + v_k = 0 \Rightarrow v_1 = v_2 = \dots = v_k = 0.$$

Proof. We prove this by contradiction. By renumbering the vectors, we can assume that $v_1 \dots v_m \neq 0$ and $v_{m+1} = v_{m+2} = \dots = v_k = 0$ for $1 \leq m \leq k$. If

$$v_1 + \dots + v_k = 0$$

then

$$v_1 + v_2 + \dots + v_m = 0$$

which is not possible because $v_1, \dots, v_m$ are eigenvectors corresponding to different eigenvalues $\lambda_1, \dots, \lambda_m$ and so must be linearly independent by Theorem 5.5. □

Theorem. [2] Let T be a linear operator on a vector space and let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of T. For each i, we let S_i be a finite linearly independent subset of E_{λ_i}. Then $S = S_1 \cup S_2 \cup \dots \cup S_k$ is a linearly independent subset of V.

[2] This is Theorem 5.5 in the textbook and *our* Theorem 5.5 does not appear anywhere as far as I can tell.

Proof. Let $S_i = \{v_{i,1}, v_{i,2}, \ldots, v_{i,n_i}$ for $1 \leq i \leq k$. Then

$$S = \bigcup_{i=1}^{k} S_i = \{v_{i,j} | 1 \leq i \leq k, 1 \leq j \leq n_i\}.$$

Assume that

$$\sum_{\substack{1 \leq i \leq k \\ 1 \leq j \leq n_i}} a_{i,j} v_{i,j} = 0 \qquad (a_{i,j} \in \mathbb{F}). \tag{1}$$

For each i, let

$$w_i = \sum_{j=1}^{n_i} a_{i,j} v_{i,j}.$$

From (1), we get

$$w_1 + w_2 + \cdots + w_k = 0.$$

Hence

$$\sum_{j=1}^{n_i} a_{i,j} v_{i,j} = 0$$

and $a_{i,j} = 0$ for $1 \leq i \leq k, 1 \leq j \leq n_i$ i.e. S is linearly independent by Theorem 5.5. □

Theorem 5.8. Let T be a linear operator on a finite dimensional vector space V such that the characteristic polynomial of T splits. Let $\lambda_1, \ldots, \lambda_k$ be all distinct eigenvalues of T. Then

(i) T is diagonalizable if and only if the algebraic multiplicity of λ_i is equal to $\dim(E_{\lambda_i})$.

(ii) If T is diagonalizable and β_i is an ordered basis for each $1 \leq i \leq k$, then

$$\beta = \beta_1 \cup \beta_2 \cup \cdots \cup \beta_k$$

is an ordered basis for V containing eigenvectors of T.

Proof. (i) ($\Rightarrow$) Suppose T is diagonalizable. By Theorem 5.1, there exists an ordered basis β consisting of eigenvectors of T. Let m_i denote that algebraic multiplicity of λ_i, $d_i = \dim(E_{\lambda_i})$ and $n = \dim(V)$ for $1 \leq i \leq k$.

Let $\beta_i = B \cup E_{\lambda_i}$. Let n_i be the number of vectors in β_i. Then $n_i \leq d_i$ and $d_i \leq m_i$ for $1 \leq i \leq k$.

Hence $\beta = \beta_1 \cup \beta_2 \cup \cdots \cup \beta_k$ and

$$n = \sum_{i=1}^{k} n_i \leq \sum_{i=1}^{k} d_i \leq \sum_{i=1}^{k} m_i = n.$$

It follows that

$$\sum_{i=1}^{k} (m_i - d_i) = 0$$

Since $m_i - d_i \in \mathbb{Z}_+$, $m_i = d_i$ for $1 \leq i \leq k$.

($\Leftarrow$) Now suppose $m_i = d_i$ for $1 \leq i \leq k$. For each i, let β_i be an ordered basis for E_{λ_i} and let $\beta = \beta_1 \cup \cdots \cup \beta_k$. By Theorem 5.5, β is linearly independent. Furthermore since $d_i = m_i, \beta$ contains

$$\sum_{i=1}^{k} d_i = \sum_{i=1}^{k} m_i = n$$

vectors. Therefore β is an ordered basis for V consisting of eigenvectors of K. Hence T is diagonalizable.

□

Example. Let T be a linear operator on $P_2(\mathbb{R})$ defined by

$$T(f(x)) = f(1) + f'(0)x + (f'(0) + f''(0))x^2 \qquad (f(x) \in P_2(\mathbb{R})).$$

Let $\alpha = \{1, x, x^2\}$ be the standard normal ordered basis for $P_2(\mathbb{R})$ and $B = [T]_\alpha$. Then

$$B = [[T(1)]_\alpha, [T(x)]_\alpha, [T(x^2)]_\alpha] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

The characteristic polynomial of B is given by

$$\det(B - tI_3) = \det \begin{bmatrix} 1-t & 1 & 1 \\ 0 & 1-t & 0 \\ 0 & 1 & 2-t \end{bmatrix} = (1-t)^2(2-t).$$

The multiplicity of $\lambda_1 = 1$ is 2 and the multiplicity of $\lambda_2 = 2$ is 1. Let's find a basis for E_{λ_1}.

$$E_{\lambda_1} = \{(x_1, x_2, x_3) | (B - I_3)(x_1, x_2, x_3) = 0\}.$$

$$B - I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

which implies $x_2 + x_3 = 0$. Hence

$$(x_1, x_2, x_3) = (x_1, 0, 0) + (0, x_2, -x_2) = x_1(1, 0, 0) + x_2(0, 1, -1).$$

Then $\{(1, 0, 0), (0, 1, -1)\}$ is a basis for E_{λ_1}. Therefore the multiplicity of λ_1 is equal to the geometric multiplicity of λ_1. Similarly, $\{(1, 0, 1)\}$ is a basis for E_{λ_2} and the geometric multiplicity of λ_2 is equal to the multiplicity of λ_2.
Hence T is diagonalizable.
Note that $\beta = \{1, x^2 - x, 1 + x^2\}$ is an eigenbasis for $P_2(\mathbb{R})$. Let

$$Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Then

$$Q^{-1}BQ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

5.3 Direct Sums

Definition. Let $W_1, \ldots, W_k$ be subspaces of a vector space V. We define the sum of these k subspaces to be the set

$$W_1 + W_2 + \cdots + W_k = \{v_1 + \cdots + v_k | v_i \in W_i\}.$$

Clearly $W_1 + \cdots + W_k$ is a subspace.

Definition. Let $W_1, \ldots, W_k$ be subspaces of a vector space V. We call V the direct sum of these k subspaces and writes

$$V = W_1 \oplus \cdots \oplus W_K = \bigoplus_{i=1}^{k} W_i$$

if

(i) $V = W_1 + \cdots + W_k$

(ii) $W_j \cap \left(\sum_{i \neq j} W_i\right) = \{0\}$.

Example. $\mathbb{R}^2 = W_1 \oplus W_2$ with $W_1 = \{(x_1, 0) : x_1 \in \mathbb{R}\}, W_2 = \{(0, x_2) : x_2 \in \mathbb{R}\}$.
$\mathbb{R}^3 = W_1 \oplus W_3$ with $W_1 = \{(x_1, x_2, 0) : x_1, x_2 \in \mathbb{R}\}$, $W_3 = \{(0, 0, x_3) : x_3 \in \mathbb{R}\}$.

Theorem 5.9. Let $W_1, \ldots, W_k$ be subspaces of a finite dimensional vector space V. Then the following are equivalent:

(i) $V = W_1 \oplus \cdots \oplus W_k$.

(ii) $V = \sum_{i=1}^{k} W_i$ and for any vectors $v_1, v_2, \ldots, v_k$ such that $v_1 \in W_i (1 \leq i \leq k)$ if

$$v_1 + v_2 + \cdots + v_k = 0$$

then $v_i = 0$ for all $1 \leq i \leq k$.

(iii) Each vector $v \in V$ can be uniquely written as

$$v_1 + \cdots + v_k$$

where $v_i \in W_i$.

(iv) If γ_i is an ordered basis for W_i then

$$\gamma_1 \cup \gamma_2 \cup \cdots \cup \gamma_k$$

is an ordered basis for V.

(v) For each $1 \leq i \leq k$, there exists an ordered basis γ_i such that

$$\gamma_1 \cup \gamma_2 \cup \cdots \cup \gamma_k$$

is an ordered basis for V.

Theorem 5.10. A linear operator T on a finite dimensional vector space V is diagonalizable if and only if V is the direct sum of the eigenspaces of T.

Proof. Let $\lambda_1, \ldots, \lambda_k$ be all the different eigenvalues of T.
($\Rightarrow$) Suppose T is diagonalizable. For each $i = 1, \ldots, k$ choose an ordered basis γ_i for the corresponding eigenspace E_{λ_i}. Then by Theorem 5.8 (ii)

$$\gamma_1 \cup \gamma_2 \cup \cdots \cup \gamma_k$$

is a basis for V and hence V is the direct sum of E_{λ_i} i.e.

$$V = \bigoplus_{i=1}^{k} E_{\lambda_k}$$

by Theorem 5.9 (v).
($\Leftarrow$) Suppose $V = \bigoplus_{i=1}^{k} E_{\lambda_k}$. Let γ_i be a basis for E_{λ_i} $(1 \leq i \leq k)$. Then

$$\gamma_1 \cup \gamma_2 \cup \cdots \cup \gamma_k$$

is a basis for V. Hence V has a basis consisting of eigenvectors of T. Therefore T is diagonalizable. $\square$

5.4 Invariant subspaces and Cayley-Hamilton Theorem

Definition. Let T be a linear operator on a finite dimensional vector space V. A subspace W is called a T-invariant subspace of V if

$$T(W) \subseteq W.$$

Where

$$T(W) = \{T(x) | x \in W\}.$$

Example. Let T be a linear operator on V. Then the following subspaces are T invariant.

(1) $\{0\}$

(2) V

(3) $R(T)$

(4) $N(T)$

(5) E_λ

Definition. Let T be a linear operator on a finite dimensional vector space V and let $x \in V$, $x \neq 0$. Then the subspace

$$W := \operatorname{span}(x, T(x), T^2(x), \dots, T^m(x), \dots)$$

is called the T-cyclic subspace of V generated by x.
Clearly T-cyclic subspaces of V generated by x are T-invariant.

Example. Let T be the linear operator on $P_2(\mathbb{R})$ defined by

$$T(f(x)) = f'(x).$$

The T-cyclic subspace generated by x^2 is

$$\operatorname{span}(x^2, T(x^2), T^2(x^2)) = \operatorname{span}(x^2, 2x, 2) = P_2(\mathbb{R}).$$

Theorem 5.20. Let T be a linear operator on a finite dimensional vector space V and let W be a T-invariant subspace. Then $T_W : W \to W$ defined by

$$T_W(x) = T(x) \qquad (x \in W)$$

is a linear operator on W and the characteristic polynomial of T_W divides the characteristic polynomial of T.

Proof. Choose an ordered basis $\gamma = \{v_1, \dots, v_k\}$ for W and extend γ to an ordered basis $\beta = \{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ for V where $\dim(V) = n, \dim(W) = k$. Then

$$A = [T]_\beta \text{ and } B_1 = [T_W]_\gamma$$

and we have

$$\begin{aligned} A &= [[T(v_1)]_\beta, \dots, [T(v_k)]_\beta, [T(v_{k+1})]_\beta, \dots, [T(v_n)]_\beta] \\ &= [[T_W(v_1)]_\beta, \dots, [T_W(v_k)]_\beta, [T_W(v_{k+1})]_\beta, \dots, [T_W(v_n)]_\beta] \end{aligned}$$

Similarly,

$$B_1 = [[T_W(v_1)]_\beta, \dots, [T_W(v_k)]_\beta]$$

So

$$A = \left[\begin{array}{c|c} B_1 & B_2 \\ \hline 0 & B_3 \end{array}\right]$$

Then

$$\det(A - tI_n) = \det \left[\begin{array}{c|c} B_1 - tI_k & B_2 \\ \hline 0 & B_3 - tI_{n-k} \end{array}\right] = \det(B_1 - tI_k)\det(B_3 - tI_{n-k}).$$

□

Theorem 5.21. Let T be a linear operator on a finite dimensional vector space V and let W be the T-cyclic subspace of V generated by a non-zero vector v. Let $k = \dim(W)$. Then

(i) $\{v, T(v), \ldots, T^{k-1}(v)\}$ is a basis for W.

(ii) If $a_0 v + a_1 T(v) + \cdots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$, then the characteristic polynomial of T_W is $f(t) = (-1)^k (a_0 + a_1 t + \cdots + a_{k-1} t^{k-1} + t^k)$.

Proof. (i) Since $v \neq 0$, $\{v\}$ is linearly independent. Choose $j \geq 1$ such that j is the largest positive integer for which

$$\beta = \{v, T(v), \ldots, T^{j-1}(v)\}$$

is linearly independent. Let $Z = \operatorname{span}(\beta)$. Then β is a basis for Z. Note $T^j(v) \in Z$ because

$$\{v, \ldots, T^{j-1}(v), T^j(v)\}$$

is linearly dependent and β is linearly independent. It follows that Z is T-invariant. Hence, we get $Z \subseteq W$ because $\beta \subseteq W$.

However we have $W \subseteq Z$ because W is the smallest T-invariant subspace which contains V. This proves $Z = W$ and so $\dim(Z) = \dim(W)$ which gives $j = k$.

(ii) Recall $T_W : W \to W$ is defined by

$$T_W(x) = T(x) \qquad (x \in W).$$

By (i) we have a basis $\beta = \{v, T(v), \ldots, T^{k-1}(v)\}$. Then we have

$$\begin{aligned}
[T_W]_\beta &= [[T_W(v)]_\beta, [T_W(T(v))])\beta, \ldots, [T_W(T^{k-1}(v))])\beta] \\
&= [[T(v)]_\beta, [T^2(v)]_\beta, \ldots, [T^k(v)]_\beta] \\
&= \begin{bmatrix} 0 & 0 & \ldots & 0 & -a_0 \\ 1 & 0 & \ldots & 0 & -a_1 \\ 0 & 1 & \ldots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & -a_{k-1} \end{bmatrix}
\end{aligned}$$

Then

$$\det([T_W]_\beta - tI) = (-1)^k (a_0 + a_1 t + \cdots + a_{k-1} t^{k-1} + t^k).$$

□

Theorem 5.22 (Cayley-Hamilton Theorem). If $f(t)$ is the characteristic polynomial of a linear operator T on a finite dimensional vector space V, then $f(T) = T_0$, i.e.

$$(f(T))(v) = 0 \qquad (v \in V). \tag{*}$$

Proof. It suffices to prove (*) for an arbitrary non-zero vector v.
Let W be the T-cyclic subspaces generated by v and suppose $\dim(W) = k$. By Theorem 5.21 (i), there exists scalars $a_0, \ldots, a_{k-1}$ such that

$$a_0 v + a_1 T(v) + \cdots + a_{k-1} T^{k-1}(v) + T^k(v) = 0 \tag{**}$$

So by Theorem 5.21 (ii), this equation implies that

$$g(t) = (-t)^k (a_0 + a_1 v + \cdots + a_{k-1} t^{k-1} + t^k)$$

is the characteristic polynomial for T_W. Hence we have

$$(g(T_W))(v) = (-1)^k (a_0 v + a_1 T_W(v) + \cdots + a_{k-1} T_w^{k-1}(v) + T_W^k(v)) = 0$$

by (**). Hence

$$(g(T))(v) = 0.$$

By Theorem 5.20, $g(t)$ divides $f(t)$. Hence

$$f(t) = g(t)h(t)$$

for some polynomial $h(t)$. Then

$$(f(T))(v) = (g(T)h(T))(v) = (h(T)g(T))(v) = 0.$$

□

Chapter 7

Canonical Forms

7.1 The Jordan Canonical Form

Definition. Let T be a linear operator on a vector space and let λ be a scalar. A non-zero vector v is called a generalized eigenvector of T corresponding to λ if

$$(T - \lambda I)^k(v) = 0$$

for some positive integer k.

Definition. Let T be a linear operator on a vector space V and let λ be an eigenvalue of T. Then, the generalized eigenspace of T corresponding to λ, denoted K_λ is the subspace of V defined by

$$K_\lambda = \{v \in V | (T - \lambda I)^p(v) = 0 \text{ for some positive integer } p\}$$

Theorem 7.1. Let T be a linear operator on a vector space V and let λ be an eigenvalue of T. Then

(i) K_λ is a T-invariant subspace of V containing the eigenspace E_λ.

(ii) For any scalars $\mu \neq \lambda$, the restriction of $T - \mu I$ is one-to-one.

Proof. (i) Clearly $0 \in K_\lambda$, hence K_λ is not empty. Let $x, y \in K_\lambda$. Then there exists positive integers p, q such that

$$(T - \lambda I)^p(x) = 0 \text{ and } (T - \lambda I)^q(y) = 0.$$

Then

$$\begin{aligned}(T - \lambda I)^{p+q}(x + y) &= (T - \lambda I)^{p+q}(x) + (T - \lambda I)^{p+q}(y) \\ &= (T - \lambda I)^q((T - \lambda I)^p(x)) + (T - \lambda I)^p((T - \lambda I)^q(y)) = 0.\end{aligned}$$

Hence $x+y \in K_\lambda$. Similarly, $ck \in K_\lambda$ for $c \in \mathbb{F}$ and $x \in K_\lambda$. Therefore K_λ is a subspace of V.

Clearly $E_\lambda \subseteq K_\lambda$.

Now let $x \in K_\lambda$. Then

$$(T - \lambda I)^p(x) = 0$$

for some p. Then

$$(T - \lambda I)^p(T(x)) = ((T - \lambda I)^p T)(x) = (T(T - \lambda I)^p(x) = T(0) = 0.$$

Hence $T(x) \in K_\lambda$.

(ii) We will prove this by contradiction. Assume $(T - \mu I)$ is not one-to-one on K_λ. Then

$$(T - \mu I)(x) = 0$$

for some $0 \neq x \in K_\lambda$. Let p be the smallest positive integer for which

$$(T - \lambda I)^p(x) = 0.$$

Let

$$y = (T - \lambda I)^{p-1}(x) \neq 0.$$

Then

$$(T - \lambda I)(y) = (T - \lambda I)((T - \lambda I)^{p-1}(x)) = (T - \lambda I)(0) = 0.$$

Hence $y \in E_\lambda$. Then

$$\begin{aligned}(T - \mu I)(y) &= (T - \mu I)((T - \lambda I)^{p-1}(x)) \\ &= ((T - \mu I)(T - \lambda I)^{p-1})(x) \\ &= (T - \lambda I)^{p-1}((T - \mu I)(x)) = 0.\end{aligned}$$

Hence $y \in E_\mu$ which is a contradiction.

□

Theorem 7.2. Let T be a linear operator on a finite dimensional vector space V such that the characteristic polynomial of T splits. Suppose λ is an eigenvalue of T with algebraic multiplicity m. Then

(a) $1 \leq \dim(K_\lambda) \leq m$.

(b) $K_\lambda = N((T - \lambda I)^m)$.

Proof. (a) Let $W = K_\lambda$, and let $h(t)$ be the characteristic polynomial of $T_W : W \to W$. By Theorem 5.20, $h(t)$ divides the characteristic polynomial of T on V and by Theorem 7.21 (b), λ is the only eigenvalue of T_W. In fact, if μ is any other eigenvalue of T_W, then $(T_W - \mu I)(w) = 0$ for some eigenvectors of T_W. Then $T - \mu I$ is not one-to-one on $W = K_\lambda$ which is a contradiction. Then implies $h(t) = (-1)^d(t - \lambda)^d$ where $d = \dim(W) = \dim(K_\lambda)$. Hence $d \leq m$ because $h(t)|f(t)$.

(b) Clearly

$$N((T-\lambda I)^m) \subseteq K_\lambda. \tag{1}$$

Conversely, let $W = K_\lambda$ and $h(t)$ be the characteristic polynomial of $T_W : W \to W$. By the Cayley-Hamilton theorem, $h(T_W) = 0$. Hence we get

$$h(T_W)(x) = 0 \qquad (x \in W).$$

By the proof of part a, we have

$$h(t) = (-1)^d (t-\lambda)^d.$$

where $d = \dim(K_\lambda)$. So

$$(-1)^d (T_W - \lambda I)^d (x) = 0.$$

and so

$$(T-\lambda I)^d(x) = (T_W - \lambda I)^d(x) = 0$$

for any $x \in W$. It follows that

$$(T-\lambda I)^m(x) = (T-\lambda I)^{m-d}(T-\lambda I)^d(x) = 0.$$

So $x \in N((T-\lambda I)^m)$ i.e.

$$K - \lambda \subseteq N((T-\lambda I)^m). \tag{2}$$

By (1) and (2), $K_\lambda = N((T-\lambda I)^m)$.

□

Theorem 7.3. Let T be a linear operator on a finite dimensional vector space V such that the characteristic polynomial of T splits and let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T. Then for every $x \in V$, there exist vectors $v_1 \in K_{\lambda_i}$ for $i = 1, \ldots, k$ such that

$$x = v_1 + v_2 + \cdots + v_k.$$

In other words $V = K_{\lambda_1} + \cdots + K_{\lambda_k}$.

Theorem 7.4. Let T be a linear operator on a finite dimensional vector space V whose characteristic polynomial splits. For $i = 1, 2 \ldots, k$, let λ_i be the distinct eigenvalues of T with algebraic multiplicities m_i. Let β_i be the ordered basis for K_{λ_i}. Then

(a) $\beta_i \cap \beta_i = \emptyset$.

(b) $\beta = \beta_1 \cup \cdots \cup \beta_k$ is a basis for V.

(c) $\dim(K_{\lambda_i}) = m_i$.

Definition. Let T be a linear operator on a finite dimensional vector space V. Let $0 \neq x$ be a generalized eigenvector of T associated to λ. Suppose that p is the smallest positive integer for which

$$(T-\lambda I)^p(x) = 0.$$

Then the ordered set

$$\{(T-\lambda I)^{p-1}(x), \ldots, (T-\lambda I)(x), x\}$$

is a cycle of generalized eigenvectors of T corresponding to λ. The vectors $(T-\lambda I)^{p-1}(x)$ and x are called the initial vector and the end vector of the cycle. We say the length of the cycle is p.

Theorem 7.5. Let T be a linear operator on a finite dimensional vector space V with characteristic polynomial that splits and suppose β is a basis for V such that β is a disjoint union of cycles of the generalized eigenvectors of T. Then the following are true:

(a) For each cycle γ of generalized eigenvectors contained in β, $W = \text{span}(()\gamma)$ is T-invariant and $[T_W]_\gamma$ is a Jordan block, i.e.

$$[T_W]_\gamma = \begin{bmatrix} \lambda & 1 & 0 & \ldots & 0 \\ 0 & \lambda & 1 & \ldots & 0 \\ 0 & 0 & \lambda & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & \lambda \end{bmatrix}$$

(b) β is a Jordan canonical basis for V, i.e.

$$[T]_\beta = \left[\begin{array}{c|c|c|c} A_1 & 0 & \ldots & 0 \\ \hline 0 & A_2 & \ldots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \ldots & A_k \end{array}\right]$$

where each A_i is a Jordan block.

Proof. (a) Suppose that the cycle γ corresponds to the eigenvalue λ, γ has length p and x is the end vector of γ. Then

$$\gamma = \{v_1, v_2, \ldots, v_p\}$$

where $v_p = x$ and $v_{p-i} = (T-\lambda)^i(x)$ for $i = 1, \ldots, p-1$. Then

$$\begin{aligned} T(v_1) &= T((T-\lambda I)^{p-1}(x)) = [(T-\lambda I)+\lambda I]((T-\lambda I)^{p-1}(x)) \\ &= (T-\lambda I)((T-\lambda I)^{p-1}(x) + \lambda I((T-\lambda I)^{p-1}(x)) \\ &= (T-\lambda I)^p(x) + \lambda(T-\lambda I)^{p-1}(x) \\ &= \lambda v_1. \end{aligned}$$

Hence

$$[T(v_1)]_\gamma = (\lambda, 0, \ldots, 0)$$

Similarly,

$$\begin{aligned} T(v_1) &= T((T-\lambda I)^{p-2}(x)) = [(T-\lambda)+\lambda](T-\lambda I)^{p-2}(x) \\ &= (T-\lambda I)((T-\lambda I)^{p-2}(x) + \lambda I((T-\lambda I)^{p-2}(x) \\ &= (T-\lambda I)^{p-1}(x) + \lambda I(T-\lambda I)^{p-2}(x) \\ &= v_1 + \lambda v_2. \end{aligned}$$

Hence
$$[T(v_2)]_\gamma = (1, \lambda, \ldots, 0).$$
For $1 < i \leq p$, then
$$[T(v_2)]_\gamma = (0, \ldots, 1, \lambda, \ldots, 0).$$
Hence $[T_W]_\gamma$ is a Jordan block.

(b) Part (b) follows from part (a).

□

Theorem 7.7. Let T be a linear operator on a finite dimensional vector space V and let λ be an eigenvalue of T. Then K_λ has an ordered basis consisting of a union of disjoint cycles of generalized eigenvectors corresponding to λ.

Note. This along with Theorem 7.4 guarantees a Jordan canonical form for all linear operators.

Example. Let
$$A = \begin{bmatrix} 3 & 1 & -2 \\ -1 & 0 & 5 \\ -1 & -1 & 4 \end{bmatrix}$$
Find the Jordan canonical basis for A.
A has the eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 2$. In fact $\det(A - tI) = (t-3)(t-2)^2$.
Then $\beta_1 = \{(-1, 2, 1)\}$ is a basis for $K_{\lambda_1} = E_{\lambda_1}$.
$\beta_2 = \{(1, -3, 1)\}$ is a basis for E_{λ_2} i.e. $E_{\lambda_2} \neq K_{\lambda_2}$.
In this case, we know there exists $x \in K_{\lambda_2}$ such that $\{(T - 2I)(x), x\}$ forms a basis for $K_{\lambda_2} = N((T - 2I)^2(x))$. Note that $\{(-1, 2, 0), (1, 0, 2)\}$ is a basis for K_{λ_2}. Hence any vector in K_{λ_2} is given by
$$a(-1, 2, 0) + b(1, 0, 2) = (-a + b, 2a, 2b).$$
We want to choose x such that $x \in K_{\lambda_2}$, $x \notin E_{\lambda_2}$.
Then
$$(T - 2I)(x) = \begin{bmatrix} 1 & 1 & -2 \\ -1 & -2 & 5 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} -a + b \\ 2a \\ 2b \end{bmatrix} = \begin{bmatrix} a - 3b \\ -3a + 9b \\ -a + 3b \end{bmatrix}$$
where $a \neq 3b$. Hence a cycle of the generalized eigenvectors corresponding to $\lambda_2 = 2$ is given by
$$\beta_2 = \{(a - 3b, -3a + 9b, -a + 3b), (-a + b, 2a, 2b)\}.$$
If we choose $a = 1, b = 0$, then
$$\beta_2 = \{(1, -3, -1), (-1, 2, 0)\}$$
which is the solution from the textbook. If we let β be the ordered basis given by $\beta_1 \cup \beta_2$, then
$$A = \begin{bmatrix} -1 & 1 & -1 \\ 2 & -3 & 2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -2 & -1 & 1 \\ -2 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

Appendix D

Complex Numbers

For the purposes of algebra, the field of real numbers is not sufficient, for there are polynomials of non zero degree with real coefficients that have no zero in the field of real numbers. It is often desirable to have a field in which any polynomial of non zero degree with coefficients from that field has a zero in that field. It is possible to "enlarge" the field of real numbers to obtain such a field.

Definition. A **complex number** is an expression of the form $z = a + bi$ where a and b are real numbers called the **real part** and **imaginary part** of z, respectively.
The **sum** and **product** of two complex numbers $z = a + bi$ and $w = c + di$ (where a, b, c and d are real numbers) are defined, respectively as follows:

$$z + w = (a + bi) + (c + di) = (a + c) + (b + d)i$$

and

$$zw = (a + bi)(c + di) = (ac - bd) + (bc + ad)i.$$

The **(complex) conjugate** of a complex number $z = a + bi$ is the complex number $a - bi$. We denote the conjugate of the complex number z by $\overline{z}$.
The inverse of $z = a + bi$, with $a^2 + b^2 \neq 0$ is given by $z^{-1} = \dfrac{a - bi}{a^2 + b^2}$

Theorem D.1 (Properties of the conjugate). Let z and w be two complex numbers. Then the following statements are true.

(a) $\overline{\overline{z}} = z$.

(b) $\overline{z \pm w} = \overline{z} \pm \overline{w}$.

(c) $\overline{zw} = \overline{z} \cdot \overline{w}$.

(d) $\overline{\left(\dfrac{z}{w}\right)} = \dfrac{\overline{z}}{\overline{w}}$ if $w \neq 0$.

(e) z is a real number if and only if $z = \overline{z}$.

Proof. Let $z = a + bi$ and $w = c + di$, where $a, b, c, d \in \mathbb{R}$.

(a) $\overline{\overline{z}} = \overline{a - bi} = a + bi = z$.

(b) $\overline{z + w} = \overline{(a + c) + (b + d)i} = (a + c) - (b + d)i = a - bi + c - di = \overline{z} + \overline{w}$.

The case $z - w$ is similar.

(c) $\overline{zw} = \overline{(ac - bd) + (bc + ad)i} = (ac - bd) - (bc + ad)i = (a - bi)(c - di) = \overline{z} \cdot \overline{w}$.

(d) Let $w \neq 0$, then

$$\overline{\left(\frac{z}{w}\right)} = \overline{\left(\frac{a + bi}{c + di}\right)} = \overline{\left(\frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}\right)} = \frac{(ac + bd) - (bc - ad)i}{c^2 + d^2} = \frac{\overline{z}}{\overline{w}}.$$

(e) ($\Rightarrow$) Let z be a real number. Then $z = a + 0i$ and $\overline{z} = a - 0i = a = z$.

($\Leftarrow$) Now let $z = \overline{z}$. Then $a + bi = a - bi$ so $a - a = 2bi$ and $b = 0$. Hence $z = a$ where a is a real number, so z is a real number.

□

Definition. Let $z = a + bi$, where $a, b \in \mathbb{R}$. The **absolute value** (or **modulus**) of z is the real numbers $\sqrt{a^2 + b^2}$. We denote the absolute value of z by $|z|$.
Observe that $z\overline{z} = |z|^2$.

Theorem D.2 (Properties of the absolute value). Let z and w be two complex numbers. Then the following are true.

(a) $|zw| = |z| \cdot |w|$.

(b) $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ with $w \neq 0$.

(c) $|z + w| \leq |z| + |w|$.

(d) $|z| - |w| \leq |z + w|$.

Proof. Let z and w be two complex numbers.

(a) By (c) of Theorem D.1, we have

$$|zw|^2 = (zw)\overline{(zw)} = (zw)(\overline{z}\overline{w}) = (z\overline{z})(w\overline{w}) = |z|^2 \cdot |w|^2.$$

By taking square roots we have proven (a).

(b) Let $v = \frac{1}{w}$. By applying (a) to zv we get $|zv| = |z| \cdot |v| = \frac{|z|}{|w|}$.

(c) For any complex number $x = a + bi$ observe that $x + \overline{x} \leq 2\,|x|$.

Taking $x = w\overline{z}$, we have

$$w\overline{z} + z\overline{w} \leq |zw| = |z|\,|w|\,.$$

Then,

$$\begin{aligned} |z+w|^2 &= (z+w)\overline{(z+w)} = z\overline{z} + w\overline{z} + z\overline{w} + w\overline{w} \\ &\leq |z|^2 + 2\,|z|\,|w| + |w|^2 = (|z| + |w|)^2. \end{aligned}$$

By taking square roots we have proven (c).

(d) From (a) and (c) it follows that

$$|z| = |(z+w) - w| \leq |z+w| + |-w| = |z+w| + |w|$$

So

$$|z| - |w| \leq |z+w|$$

proving (d).

□

The motivation for expanding our field from the real numbers to complex numbers is to obtain a field such that every polynomial with non zero degree having coefficients in that field has a zero, which is a property satisfied by the complex field and is formulated more carefully in the following theorem.

Theorem D.3 (The Fundamental Theorem of Algebra). Suppose that

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + ... + a_1 z + a_0$$

is a polynomial in $P(\mathbb{C})$ of degree $n \geq 1$. Then $p(z)$ has a zero.

The following important corollary is a consequence of Theorem D.3 and the division algorithm for polynomials.

Corollary D.3.1. If $p(z) = a_n z^n + a_{n-1} z^{n-1} + ... + a_1 z + a_0$ is a polynomial of degree $n \geq 1$ with complex coefficients, then there exists complex numbers $c_1, c_2, .., c_n$ (not necessarily distinct) such that

$$p(z) = a_n (z - c_1)(z - c_2)...(z - c_n).$$

Appendix G

Topics Covered in Final Exam

- Vector spaces of $\mathbb{R}, \mathbb{Q}, \mathbb{C}$.
- Subspaces, linearly (in)dependent sets, basis, dimensions.
- Linear transformations and matrices. Nullspace, Range, Dimension/Rank-Nullity Theorem 2.3, product of two linear transformations, matrix of linear transformations $[T]_\beta^\gamma$, invertability, isomorphisms,Theorem 2.6
- Dual space.
- Theorem 2.23 Change of coordinate matrix $[T]'_\beta = Q^{-1}[T]_\beta Q$.
- Determinants.
- Diagonalization.
- Diagonalizability criteria, Theorem 5.8.
- Generalized eigenspace.
- Theorem 7.2 $K_\lambda = N((T - \lambda I)^m)$.
- Cycles of generalized eigenvectors.
- Cayley-Hamilton Theorem 5.22.

The exam will have 7 question.s The first question will be true or false. The last question will be a short proof which should take about 1/3 of the page.